Presentation 3

STAT 390 | Group 2 | Winter 2026

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Part 1: Programmers

Roma & Paige & Charlene

Mandatory Changes

Remove dropout from the attention MLP

- Original AttentionPool attention network: Linear → Tanh → **Dropout** → Linear

- Removed nn.Dropout(dropout) from the attention nn.Sequential block
- Dropout is still applied in patch_projector and classifier — only the attention MLP was changed

- Model benefit: dropout in attention layers can randomly zero out attention scores during training, destabilizing which patches the model focuses on

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Roma/models.py

Use gated attention

- Replace AttentionPool with GatedAttentionPool
  - Original AttentionPool: single branch — softmax(w(tanh(Vx)))
  - New GatedAttentionPool: two branches — softmax(w(tanh(Vx) × sigmoid(Ux)))
- Swap AttentionPool(...) to GatedAttentionPool(...) in self.patch_attention, self.stain_attention, and self.case_attention in HierarchicalAttnMIL.__init__

- Model benefit: gated attention is more expressive than single-branch attention, so it can capture more complex patterns

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Roma/models.py

Use gated attention

```python
class GatedAttentionPool(nn.Module):

    def __init__(self, input_dim: int, hidden_dim: int = 128):
        super().__init__()
        self.V = nn.Linear(input_dim, hidden_dim)
        self.U = nn.Linear(input_dim, hidden_dim)
        self.w = nn.Linear(hidden_dim, 1)

    def forward(self, x: torch.Tensor, return_weights: bool = False, return_entropy: bool = False):
        # x: (B, M, D)
        Vx = torch.tanh(self.V(x))
        Ux = torch.sigmoid(self.U(x))
        weights = torch.softmax(self.w(Vx * Ux), dim=1)  # (B, M, 1)
        weighted_x = (weights * x).sum(dim=1)             # (B, D)

        entropy = None
        if return_entropy:
            p = weights.squeeze(-1)                            # (B, M)
            entropy = -(p * torch.log(p + 1e-8)).sum(dim=1)  # (B,)

        if return_weights and return_entropy:
            return weighted_x, weights.squeeze(-1), entropy
        elif return_weights:
            return weighted_x, weights.squeeze(-1)
        elif return_entropy:
            return weighted_x, entropy
        return weighted_x
```

```python
# Gated attention modules (no dropout inside attention MLP)
self.patch_attention = GatedAttentionPool(embed_dim, MODEL_CONFIG["attention_hidden_dim"])
self.stain_attention = GatedAttentionPool(embed_dim, MODEL_CONFIG["attention_hidden_dim"])
self.case_attention  = GatedAttentionPool(embed_dim, MODEL_CONFIG["attention_hidden_dim"])
```

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Roma/models.py

Add entropy regularization for attention

- Add entropy term to the training loss: Loss = CE Loss + $\lambda \times$ Entropy Loss
- Entropy is computed and summed at all three levels: patch, slice, and case

- Model benefit: entropy measures how spread out attention weights are across patches/slices, and without regularization, attention can collapse onto just one or two patches

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Roma/models.py

Add entropy regularization for attention

- Added return_entropy parameter to GatedAttentionPool.forward
- process_single_stain now also initializes patch_entropies = []
- stain_attention_info now includes "slice_entropy" and "patch_entropies" keys
- forward now retrieves case_entropy from case_attention and includes it in the returned all_weights dict

```
entropy = None
if return_entropy:
    p = weights.squeeze(-1)                              # (B, M)
    entropy = -(p * torch.log(p + 1e-8)).sum(dim=1)    # (B,)
```

```
stain_attention_info = {
    "slice_weights":   stain_weights.squeeze(0).detach(),
    "patch_weights":   slice_attention_weights,
    "slice_entropy":   stain_entropy.squeeze(0),
    "patch_entropies": patch_entropies,
}
```

```
if return_attn_weights:
    slice_emb, patch_weights, patch_entropy = self.patch_attention(
        patch_embeddings.unsqueeze(0), return_weights=True, return_entropy=True
    )
```

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Roma/models.py

Add entropy regularization for attention

- Define self.entropy_lambda in MILTrainer __init__
- Update _forward_one_case to accept return_attn_weights=False parameter
  - Returns (logits, label, attn_info) when True
- Entropy is accumulated from all three levels: attn_info["case_entropy"], info["slice_entropy"], and info["patch_entropies"]
- Final loss computed as loss = ce_loss + self.entropy_lambda * entropy_loss

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Roma/trainer.py

Add entropy regularization for attention

```
# Entropy regularization lambda (set to 0.0 to disable)
self.entropy_lambda = TRAINING_CONFIG.get("entropy_lambda", 0.0)
```

```
entropy_loss = 0.0
if use_entropy and attn_info is not None:
    # Case-level entropy (across stains)
    if "case_entropy" in attn_info:
        entropy_loss += attn_info["case_entropy"].mean()
    # Stain & patch level entropy
    for stain_name in attn_info.get("stain_weights", {}):
        info = attn_info["stain_weights"][stain_name]
        if "slice_entropy" in info:
            entropy_loss += info["slice_entropy"].mean()
        for patch_entropy in info.get("patch_entropies", []):
            entropy_loss += patch_entropy.mean()

loss = ce_loss + self.entropy_lambda * entropy_loss
```

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Roma/trainer.py

Add entropy regularization for attention

- Added entropy_lambda to TRAINING_CONFIG

- Controls the weight of the entropy term in the loss
    - Small value so it encourages attention diversity without overwhelming the classification signal

```
TRAINING_CONFIG = {
    'epochs': 30,  # Increased since we have early
    'batch_size': 1,  # MIL typically uses batch_s
    'learning_rate': 2.08e-4,  # Updated from OPTU
    'weight_decay': 3.05e-4,  # Updated from OPTUN
    'num_workers': 2,
    'pin_memory': True,
    'random_state': 42,
    'class_weights': [2.5285, 1.0],  # Increased b
    'dropout': 0.2599,  # Add dropout for regulari
    # Learning rate scheduler
    'use_scheduler': True,
    'scheduler_type': 'reduce_on_plateau',  # 'red
    'scheduler_patience': 4,  # For ReduceLROnPlat
    'scheduler_factor': 0.2139,  # Reduce LR by ha
    'scheduler_min_lr': 1e-6,
    # Early stopping
    'early_stopping': True,
    'early_stopping_patience': 8,  # Stop if no im
    'early_stopping_min_delta': 0.001,  # Minimum
    'early_stopping_min_epochs': 30,  # Minimum ep
    'entropy_lambda': 1e-5, # Entropy regularizati
}
```

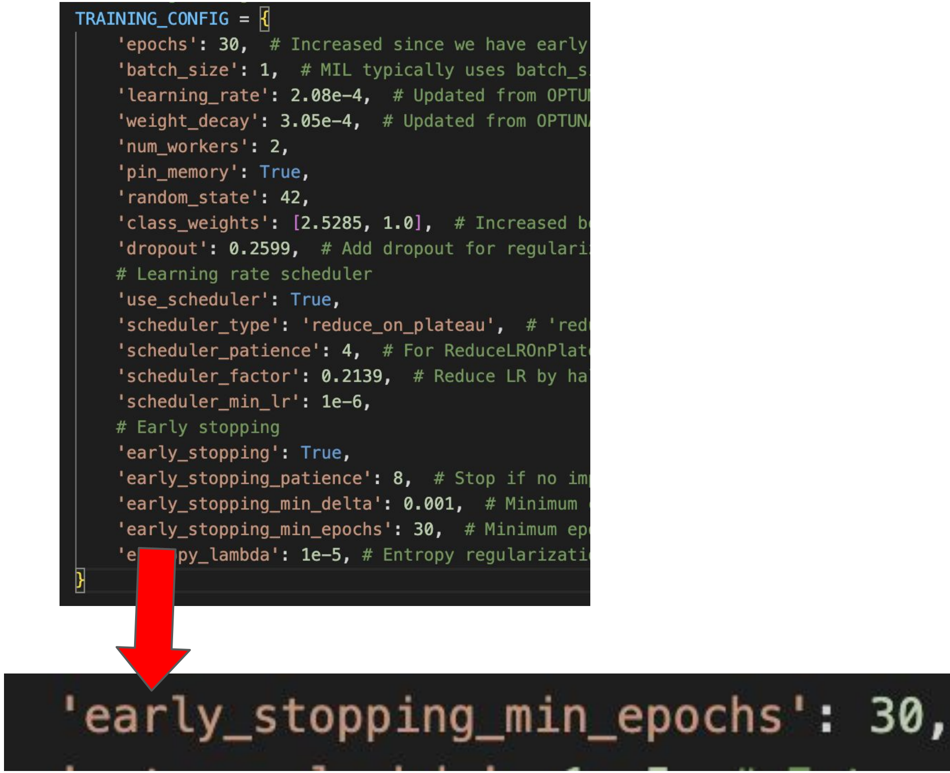
```
'entropy_lambda': 1e-5,
```

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Roma/config.py

Change minimum epochs to 30

- Early stopping minimum epochs defines how many epochs must pass before early stopping is even allowed to trigger
- Change early_stopping_min_epochs value to 30 in TRAINING_CONFIG

- Model benefit: increasing to 30 ensures the model trains for a substantial period before any early stopping decision is made



```
TRAINING_CONFIG = {
    'epochs': 30,  # Increased since we have early
    'batch_size': 1,  # MIL typically uses batch_s
    'learning_rate': 2.08e-4,  # Updated from OPTU
    'weight_decay': 3.05e-4,  # Updated from OPTUN
    'num_workers': 2,
    'pin_memory': True,
    'random_state': 42,
    'class_weights': [2.5285, 1.0],  # Increased b
    'dropout': 0.2599,  # Add dropout for regulari
    # Learning rate scheduler
    'use_scheduler': True,
    'scheduler_type': 'reduce_on_plateau',  # 'red
    'scheduler_patience': 4,  # For ReduceLROnPlat
    'scheduler_factor': 0.2139,  # Reduce LR by ha
    'scheduler_min_lr': 1e-6,
    # Early stopping
    'early_stopping': True,
    'early_stopping_patience': 8,  # Stop if no im
    'early_stopping_min_delta': 0.001,  # Minimum
    'early_stopping_min_epochs': 30,  # Minimum ep
    'e   py_lambda': 1e-5, # Entropy regularizati
}
```

```
'early_stopping_min_epochs': 30,
```

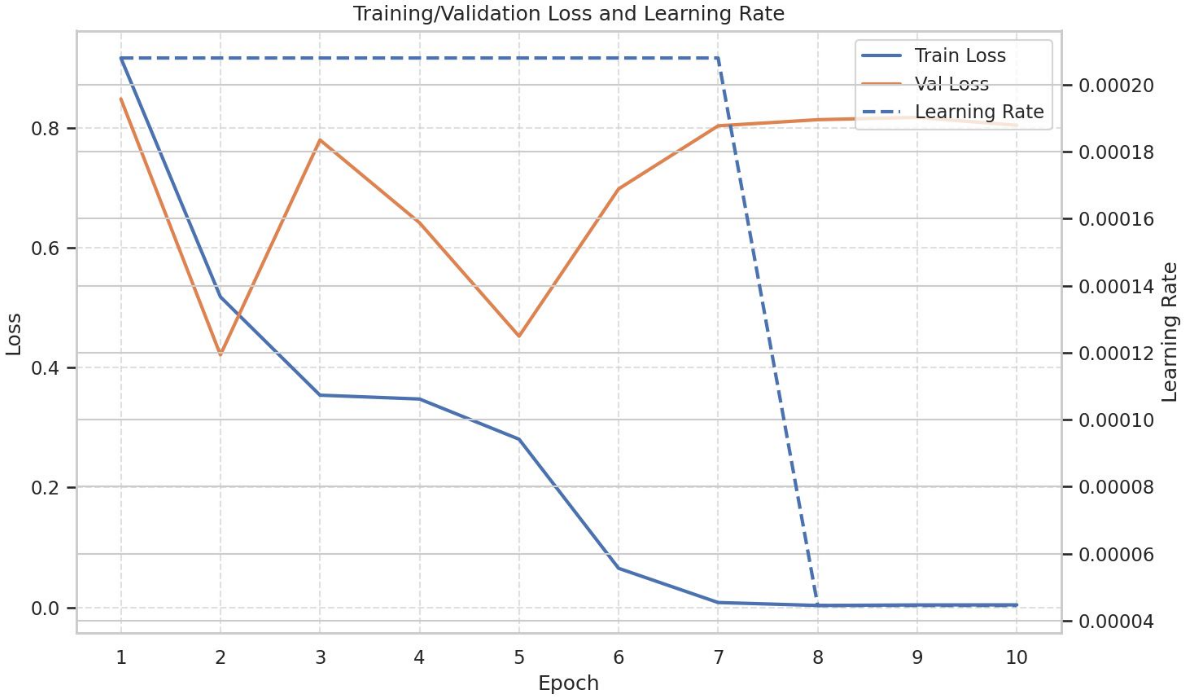
/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Roma/config.py

Use new splits

```
python3 main.py \
    --analyze_attention \
    --attention_top_n 8 \
    --per_slice_cap 500 \
    --max_slices_per_stain 5 \
    --load_splits /projects/e32998/STAT390_Krish/Code/Code4_reduce_runtime/data_splits/data_splits.npz
```

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Roma/sbatch_files/run_training_custom_split.sbatch

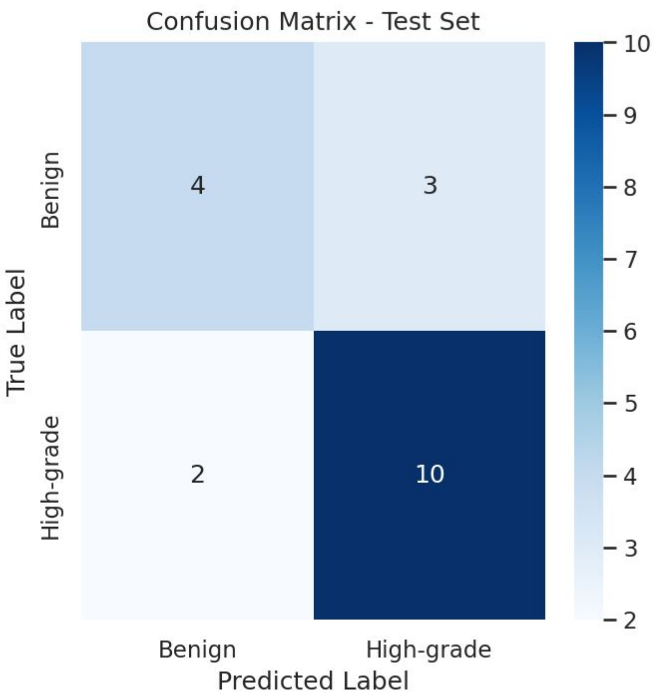
Train/Validation Loss



- Training loss decreases steadily to near zero
- Validation loss improves early (≈ epoch 2) then increases
- Large gap develops between training and validation loss
- Best validation loss: 0.4213 (early epoch)
- Early stopping triggered at epoch 10

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Charlene_Visualization/Presentation3_visualization.ipynb

Confusion Matrix



- Overall accuracy: 73.7% (14/19 correct)
- High Grade detection: 10/12 correctly identified (83%)
- Benign detection: 4/7 correctly identified (57%)
- Improvement from before
- Model is still performing better at detecting High Grade than Benign

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Charlene_Visualization/Presentation3_visualization.ipynb

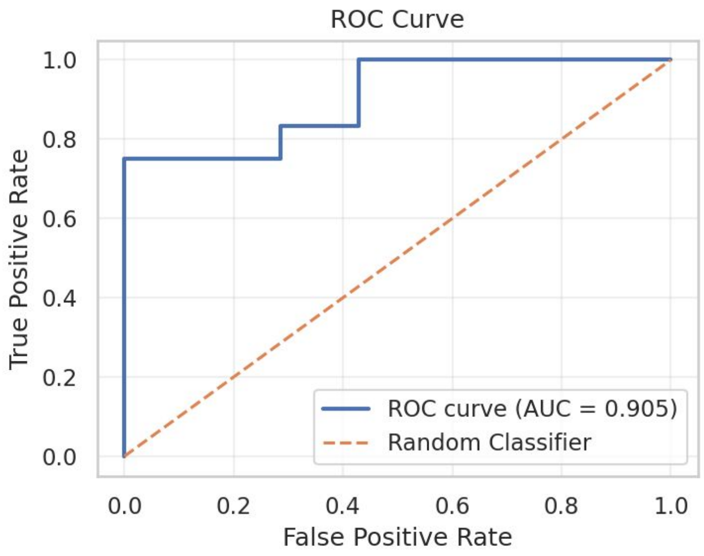
Deviation Plot

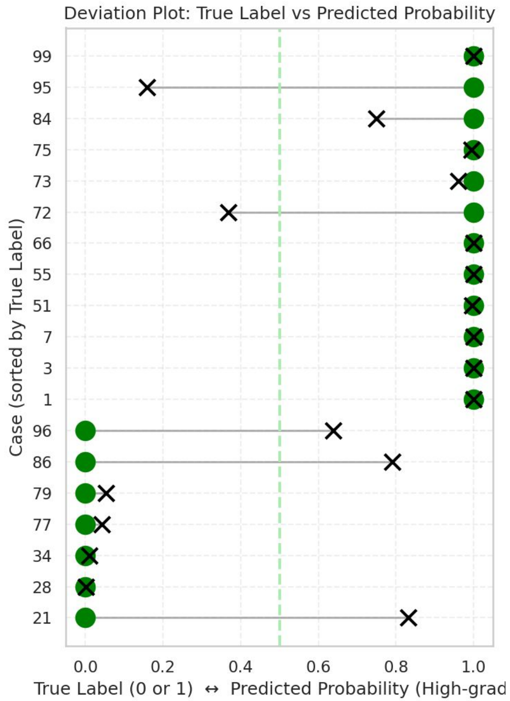
ROC Curve

- AUC = 0.905
- Strong class separation
- Performs much better than random

Deviation Plot

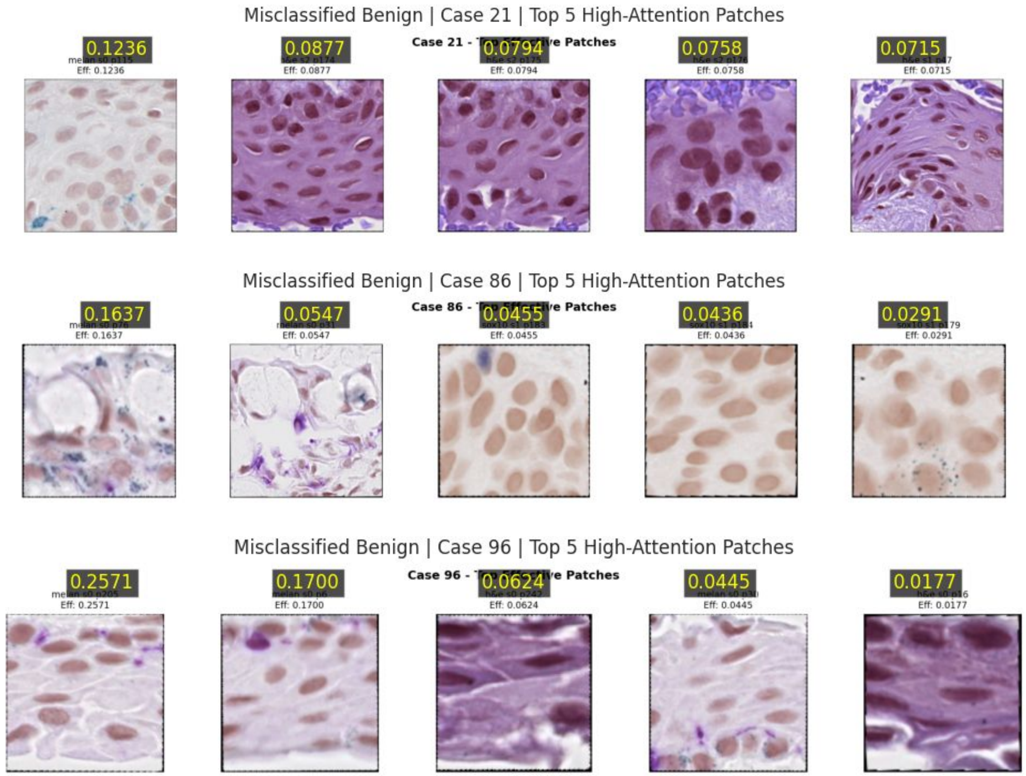
- Most predictions close to 0 or 1
- Few misclassifications near threshold
- Model is generally confident





/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Charlene_Visualization/Presentation3_visualization.ipynb

Misclassified Benign Cases



Observation

- High attention on cytologic features
- Dark, crowded, enlarged nuclei
- Confident false positives

Implication

- Model relies heavily on nuclear morphology
- Limited architectural/context awareness
- Feature overlap between benign & malignant

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Charlene_Visualization/Presentation3_visualization.ipynb

Next Steps

- Review and augment training data with benign patches showing nuclear crowding.
- Incorporate architectural context into model input or features
- Analyze model reliance on nuclear vs architectural features using attention maps.

Patch-Level Predictions

Optional Task 3

Goal *Obtaining Patch-Level Predictions*

The existing MIL model makes case-level predictions, not patch-level, which is the medically correct method. However, we want to know more surrounding the benign cases that the model is misclassifying as high-grade.

- **Motivating Question** - do some of these misclassified benign cases have a small number of "important" patches that drives the wrong prediction, even when the majority of patches within a case look benign to the model?
- Therefore, the idea is to get patch-level predictions and identify such cases - false positive cases with a minority of high-grade patches

Approach *Applying Classifier Weights to Patch Embeddings*

The existing pipeline (data loading, splitting, MIL training, and case-level evaluation) occurs just as before, unedited.

- During training the model learns:
  - patch_projector: compresses the 4096 → 512 dimensional
  - classifier: linear layer with learned beta coefficients that maps the 512-dimensional embeddings to class logits
- Normally, the patches get projected and then aggregated via attention up to the case level. Individual patches don't receive their own prediction during training.

Approach *Applying Classifier Weights to Patch Embeddings Cont.*

- We are making an **additional** post-hoc analysis to what the trained model would predict looking at just individual patches at a time.

4096-dimensional embedding → **patch_projector** → *512 dimensional* → **classifier** → *logits* → **softmax** → *probabilities*

- Doing this **after** training is complete with the learned coefficients
- For each test case, reporting:
  - True label, case-level prediction
  - % of patches predicted benign vs high-grade
- Allows us to identify benign cases where a minority of high-grade patches may have driven an incorrect case classification

Code *models.py*

The HierarchicalAttnMIL model already contains patch_projector and classifier as trained layers, but they just needed to be made accessible to call outside of the forward pass.

- One new method added: **predict_patches()**
  - Takes raw patch embeddings and runs them through patch_projector and then classifier, returning class probabilities
  - Uses same weights learned during MIL training
  - Skips attention layers

Called by trainer.py after training is complete

```
def predict_patches(self, patch_embeddings: torch.Tensor) -> torch.Tensor:
    """
    Takes (N, 4096) embeddings --> patch_projector --> (N, 512) --> classifier --> logits --> softmax -->probs
    """
    projected = self.patch_projector(patch_embeddings)  # (N, 4096) --> (N, 512)
    logits = self.classifier(projected) # (N, 512) --> (N, 2)
    return torch.softmax(logits, dim=-1) # (N, 2) probabilities
```

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Paige_Optional/models.py

Code *trainer.py*

evaluate_patches(): the core inference method that iterates over every patch in the test set, loads its embedding, and gets a patch-level prediction from the trained classifier.

- Takes test_case_dict directly from prepare_data()
- Loops case → stain → slice → patch, loads each .pt embedding file
- Calls models.predict_patches() on each individual patch
- Records case_id, true_label, probabilities, and prediction for every patch
- Saves patch_predictions.csv (one row per patch)

Pseudocode

```
evaluate_patches(test_case_dict, label_map, embeddings_dir):
    For each test case:
        For each stain → slice → patch:
            1. Load 4096-dim embedding .pt file
            2. Call models.predict_patches(embedding)
                 ●  patch_projector: 4096 → 512
                 ●  classifier:  512 → logits
                 ●  softmax: logits → class probabilities
            3. Threshold at 0.5 to get patch prediction
            4. Record: case_id, patch_path, true_label, prediction, probabilities
    Save into patch_predictions.csv (one row per patch)
```

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Paige_Optional/trainer.py

Code *trainer.py cont.*

Three supporting methods that save, summarize, and filter the patch predictions that are produced by evaluate_patches().

save_patch_predictions_csv():

- Saves all predictions and probabilities for each patch to patch_predictions.csv

```python
def _save_patch_predictions_csv(self, results: Dict[str, Any], output_dir: str,) -> str:
    self._ensure_dir(output_dir)
    df = pd.DataFrame({
        "case_id":        results["case_ids"],
        "patch_path":     results["patch_paths"],
        "true_label":     results["true_labels"],
        "predicted":      results["predictions"],
        "prob_benign":    [float(p[0]) for p in results["probs"]],
        "prob_highgrade": [float(p[1]) for p in results["probs"]],
    })
    csv_path = os.path.join(output_dir, "patch_predictions.csv")
    df.to_csv(csv_path, index=False)
    print(f"Patch predictions saved to: {csv_path}")
    return csv_path
```

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Paige_Optional/trainer.py

Code *trainer.py cont.*

Three supporting methods that save, summarize, and filter the patch predictions that are produced by evaluate_patches().

<u>Pseudocode</u>

```
summarize_patch_predictions_by_case(patch_results, case_predictions):
    For each patch result:
        Group patch predictions by case_id
    For each case:
        pct_highgrade = (# patches predicted high-grade / total patches) × 100
        pct_benign  = 100 - pct_highgrade

        Record: case_id, true_label, case-level predicted_label,
                total_patches, pct_benign, pct_highgrade

    Save into patch_summary_by_case.csv (one row per case)
```

summarize_patch_predictions_by_case():

- Aggregates back to case level by grouping patch results by case_id and computing % patches predicted benign vs high-grade
- Joins with case-level predictions from trainer.evaluate() so each row shows true label, predicted label, and patch percentages
- Saves patch_summary_by_case.csv

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Paige_Optional/trainer.py

Code *trainer.py cont.*

Three supporting methods that save, summarize, and filter the patch predictions that are produced by evaluate_patches().

identify_misclassified_benign_cases()

- Filters to cases of interest, creating a summary of cases where true_label=0 and predicted_label=1
- Saves to misclassified_benign_cases.csv

Code *main.py*

- Patch inference is added as a final post-hoc step after all existing code completes
- After trainer.train() and trainer.evaluate() complete calls:
  - evaluate_patches()

    → summarize_patch_predictions_by_case()

    →identify_misclassified_benign_cases()

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Paige_Optional/trainer.py
/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Paige_Optional/main.py

Results

Patch-level predictions aggregated back to the case level for all 19 test cases.

patch_summary_by_case

case_id	true_label	predicted_label	total_patches	pct_benign	pct_highgrade
1	1	1	452	21.46	78.54
3	1	1	862	30.51	69.49
7	1	1	502	43.63	56.37
21	0	1	710	90.85	9.15
28	0	0	1052	85.93	14.07
34	0	0	588	71.43	28.57
51	1	1	750	62.93	37.07
55	1	1	899	46.27	53.73
66	1	1	762	34.38	65.62
72	1	0	438	71.46	28.54
73	1	1	1797	65.72	34.28
75	1	1	724	69.75	30.25
77	0	0	348	85.92	14.08
79	0	0	1328	78.39	21.61
84	1	1	1181	63.59	36.41
86	0	1	1266	55.92	44.08
95	1	0	381	63.25	36.75
96	0	1	869	74.11	25.89
99	1	1	245	52.65	47.35

→ Filtered

misclassified_benign_cases

case_id	true_label	predicted_label	total_patches	pct_benign	pct_highgrade
21	0	1	710	90.85	9.15
86	0	1	1266	55.92	44.08
96	0	1	869	74.11	25.89

Case 21 is of particular interest:

- 90.85% of patches correctly predicted benign at the patch level
- Yet the case is still misclassified as high-grade

These results can then allow us to:

- filter patch_predictions.csv for Case 21's high-grade predicted patches
- Find their corresponding patch images for visual inspection
- Cross-reference with attention weights- were these patches the model assigned high attention to?

/projects/e32998/MIL_training/final_runs/run_20260228_231215/misclassified_benign_cases.csv
/projects/e32998/MIL_training/final_runs/run_20260228_231215/patch_summary_by_case.csv
/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Paige_Optional/paige_sbatch_runs/sbatch_patch

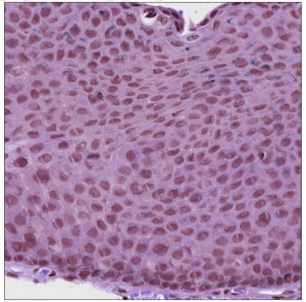
Case 21

```
Total patches in case 21: 710
High-grade predicted patches: 65
Percentage high-grade: 9.15 %
```

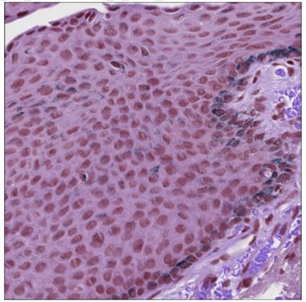
```
High-grade probability statistics:
count    65.000000
mean      0.734430
std       0.158775
min       0.507536
25%       0.595778
50%       0.714756
75%       0.866505
max       0.999999
Name: prob_highgrade, dtype: float64
```

```
Top 5 most confident high-grade patches:
                          patch_name  prob_highgrade
2523  case_21_unmatched_3_h&e_patch97.png        0.999999
2520  case_21_unmatched_3_h&e_patch94.png        0.999998
2522  case_21_unmatched_3_h&e_patch96.png        0.999984
2521  case_21_unmatched_3_h&e_patch95.png        0.999972
2524  case_21_unmatched_3_h&e_patch98.png        0.999516
```

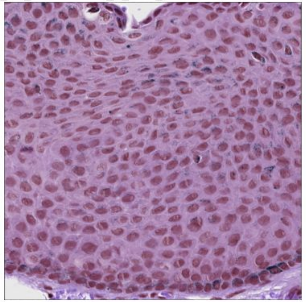
case_21_unmatched_3_h&e_patch97.png | Prob: 1.000



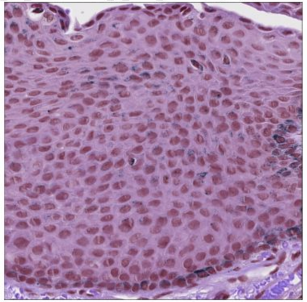
case_21_unmatched_3_h&e_patch94.png | Prob: 1.000



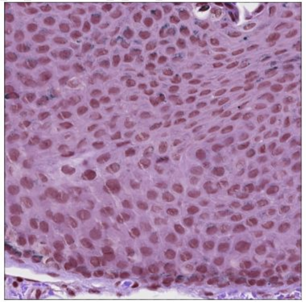
case_21_unmatched_3_h&e_patch96.png | Prob: 1.000



case_21_unmatched_3_h&e_patch95.png | Prob: 1.000



case_21_unmatched_3_h&e_patch98.png | Prob: 1.000



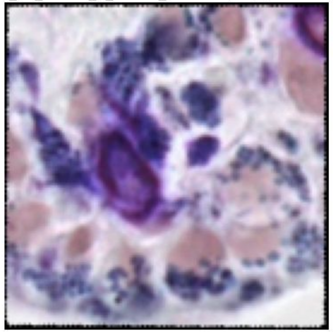
/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Charlene_Visualization/Presentation3_visualization.ipynb

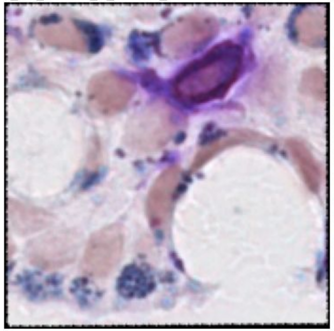
Case 86

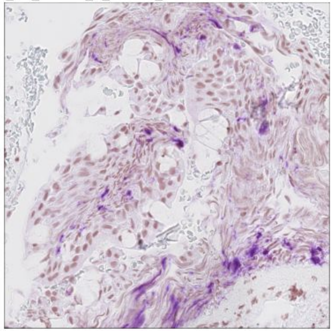
```
Total patches: 1266
High-grade predicted patches: 558
Percentage high-grade: 44.08 %
```

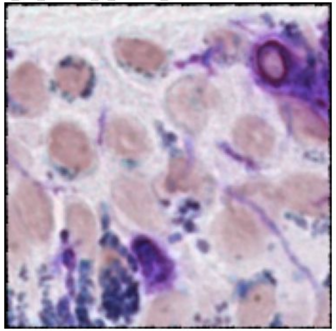
```
High-grade probability statistics:
count    558.000000
mean       0.812157
std        0.161283
min        0.500235
25%        0.672131
50%        0.841078
75%        0.971204
max        0.999995
Name: prob_highgrade, dtype: float64
```

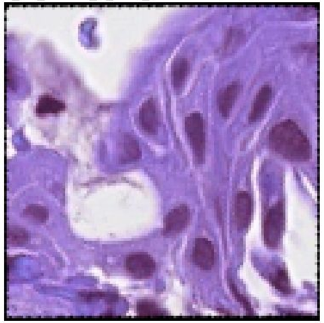
```
Top 5 most confident high-grade patches:
                          patch_name  prob_highgrade
13243   case_86_match_2_melan_patch56.png        0.999995
13207   case_86_match_2_melan_patch23.png        0.999994
12990  case_86_match_1_melan_patch167.png        0.999993
13171   case_86_match_2_melan_patch17.png        0.999993
12542    case_86_match_2_h&e_patch140.png        0.999992
```


case_86_match_2_melan_patch56.png | Prob: 1.000


case_86_match_2_melan_patch23.png | Prob: 1.000


case_86_match_1_melan_patch167.png | Prob: 1.000


case_86_match_2_melan_patch17.png | Prob: 1.000


case_86_match_2_h&e_patch140.png | Prob: 1.000

/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Charlene_Visualization/Presentation3_visualization.ipynb

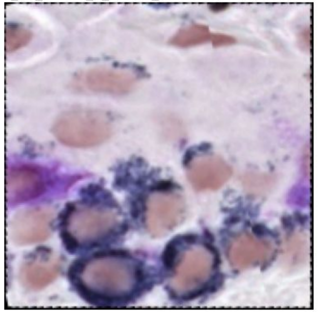
Case 96

```
Total patches: 869
High-grade predicted patches: 225
Percentage high-grade: 25.89 %
```

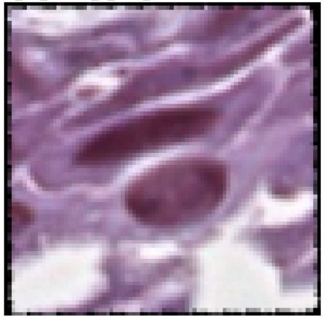
```
High-grade probability statistics:
count    225.000000
mean       0.765276
std        0.154942
min        0.501046
25%        0.627814
50%        0.758726
75%        0.913179
max        0.998875
Name: prob_highgrade, dtype: float64
```

```
Top 5 most confident high-grade patches:
                          patch_name  prob_highgrade
14638    case_96_match_1_melan_patch80.png        0.998875
14213     case_96_match_1_h&e_patch254.png        0.998866
14214     case_96_match_1_h&e_patch255.png        0.997908
14425  case_96_unmatched_1_h&e_patch92.png        0.997904
14642    case_96_match_1_melan_patch84.png        0.996503
```

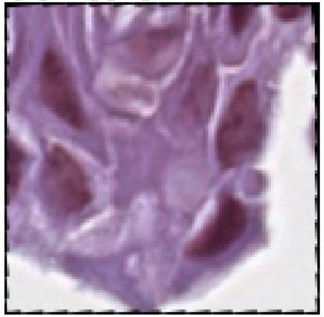
case_96_match_1_melan_patch80.png | Prob: 0.999



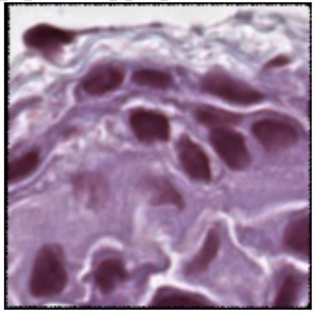
case_96_match_1_h&e_patch254.png | Prob: 0.999



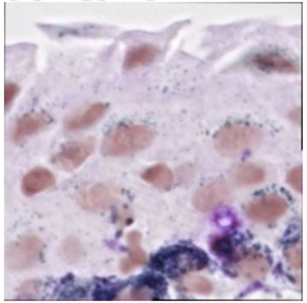
case_96_match_1_h&e_patch255.png | Prob: 0.998



case_96_unmatched_1_h&e_patch92.png | Prob: 0.998



case_96_match_1_melan_patch84.png | Prob: 0.997



/projects/e32998/STAT390_Winter2026/Team2/Code/Code_Pres3_Charlene_Visualization/Presentation3_visualization.ipynb

Next Steps

- Compare attention scores with predicted high-grade probabilities
  - Identify source of error
  - Visual inspection
- Weight patch predictions by their attention weights from the MIL model
- Run the patch-level inference on validation set also to see if pattern exists across splits
- Find out if certain stains (or other factors) produce more of the misclassified high-grade patches

Part 2: Biological Analysis

Adithi & Jason

Task & Purpose

1. Familiarize ourselves with common features of benign and high-grade slices from the Winter 2025 presentations

2. Determine which slices may be easier or harder to classify based on the features

3. Form a conclusion about the existence of certain features, or aspects of common structures, that may help distinguish benign from high-grade cases

2.1 Common Histological Patterns/Architectural Atypia

Maturation - appearance of cells as they travel deeper into the intraepithelial space

- Benign Nevi: smaller, more organized cell structure with regular spacing
- Atypia / High-Grade C-MIL: large, disorganized cell structure with clustering

Cysts

- Benign: typically have cysts
- Atypia / High-Grade C-MIL & Melanoma: typically lack cysts as lesions are more dense

Goblet Cell Hyperplasia - <u>benign</u> condition involving high counts of goblet cell mucins

2.2 Cytological Atypia Features

Nuclear Enlargement - larger nucleus, indicator of atypical cellular development

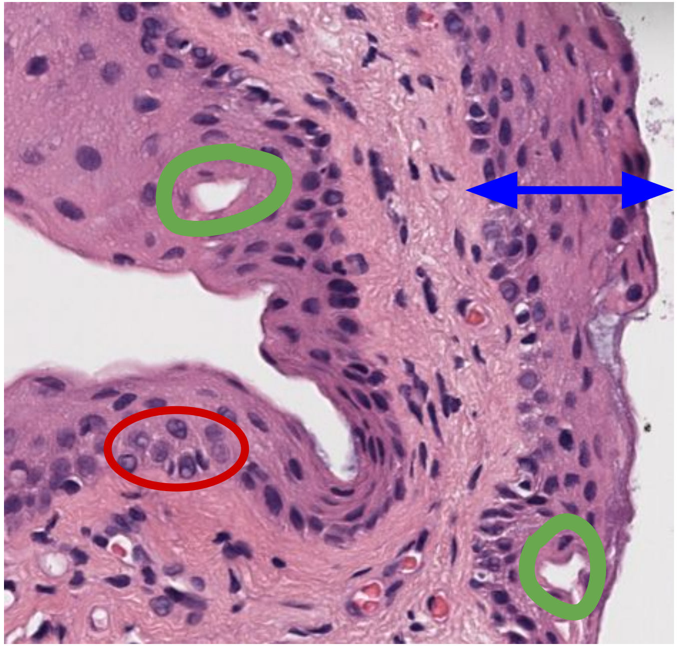
Nuclear Hyperchromasia - darker pigmented nucleus resulting from higher DNA concentration

Increased Nuclear-Cytoplasmic Ratio - nucleus takes up more space as compared to the remaining cytoplasm in the cell

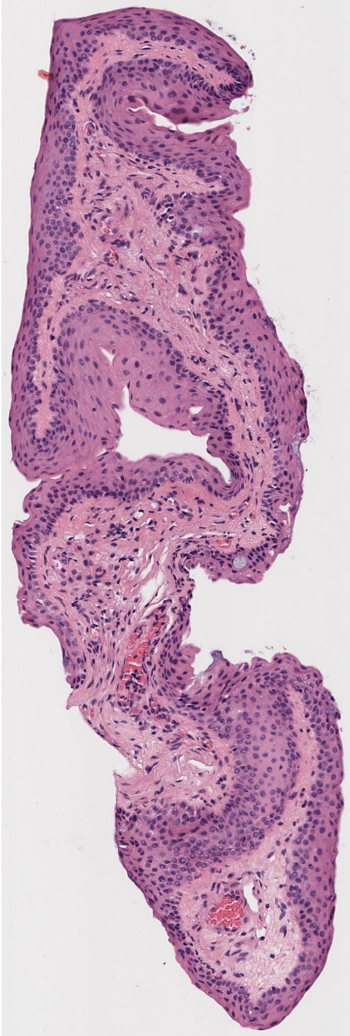
Abundant Cytoplasm - excess cytoplasm in the cell can reflect degeneration of the cell or its internal contents

Cell Structure - large, oblong cells

Case 2 - *benign*



- Cysts - benign indicator
- Maturation - benign, cells are organized
- Negligible nuclear hyperchromasia

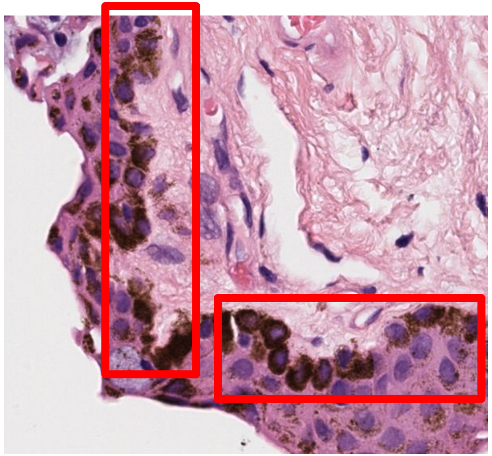


Easy to classify

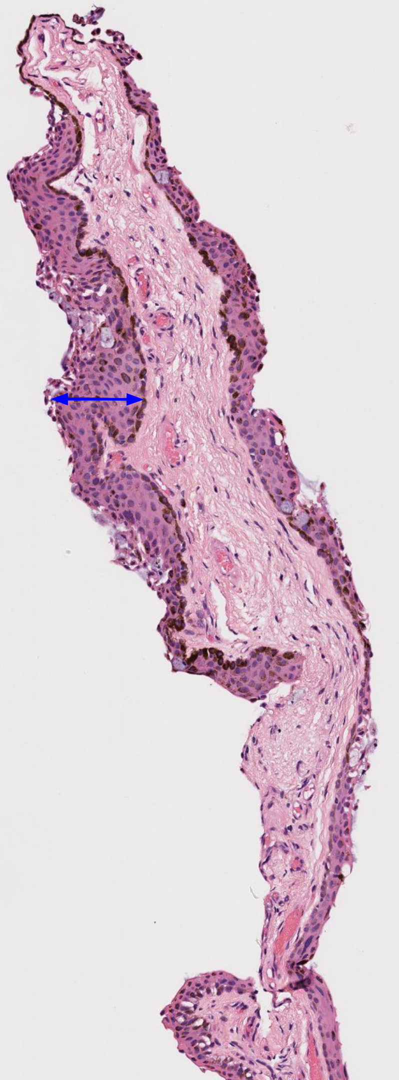
- No signs of atypia: cells in the epithelium are uniformly round & small in size
- Some goblet cell presence, but not hyperplasia
- No melanosis

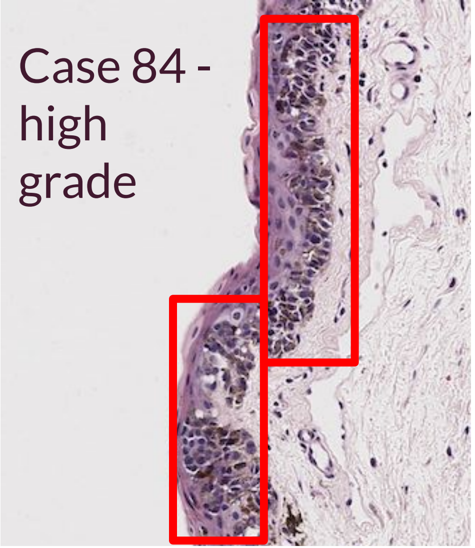
Case 4 - *benign*

<u>Hard</u> to classify



- Goblet cell hyperplasia - benign
- Cysts - benign indicator
- Basal layer melanosis
- Benign maturation (cells are organized, small, round)
- <u>Negligible atypia</u> (very few hyperchromatic nuclei along basal layer)





Disclaimer: Why We Look at Multiple Stains in Hard-to-Classify Cases

H&E as the First Step - Provides the overall tissue architecture, including epithelial layering and organization

- Highlights pigment (melanosis) and broad structural features (e.g. cysts and goblet cells)
- Serves as the initial screening stain, but pigment alone can't distinguish benign melanosis from malignant melanocytic proliferation

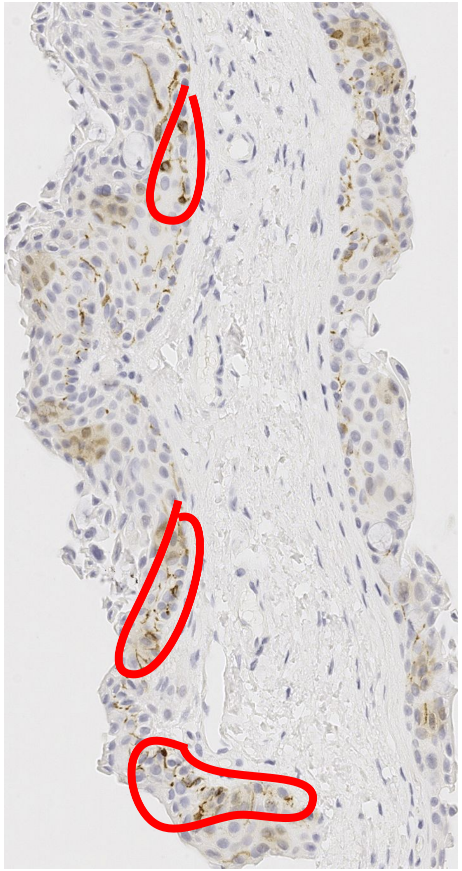
Thus, Melan-A as step 2 for Melanocyte Identification:

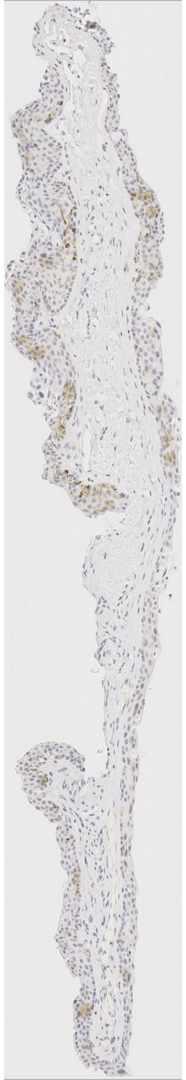
- Specifically stains melanocyte cell bodies, confirming whether pigment corresponds to melanocytes
- Helps assess melanocyte density and basal layer crowding
- Useful for distinguishing benign pigment deposition (e.g. pigment transfer to keratinocytes or other epithelial cells after production by melanocytes) from true melanocytic proliferation (atypical)

SOX10 use as step 3 for Melanocyte Distribution and Atypia:

- Nuclear stain that highlights melanocyte nuclei, making cell location clearer
- Particularly helpful for identifying suprabasal spread and subtle upward migration
- Supports evaluation of architectural disorder and early atypia

Case 4 *Melan-A*

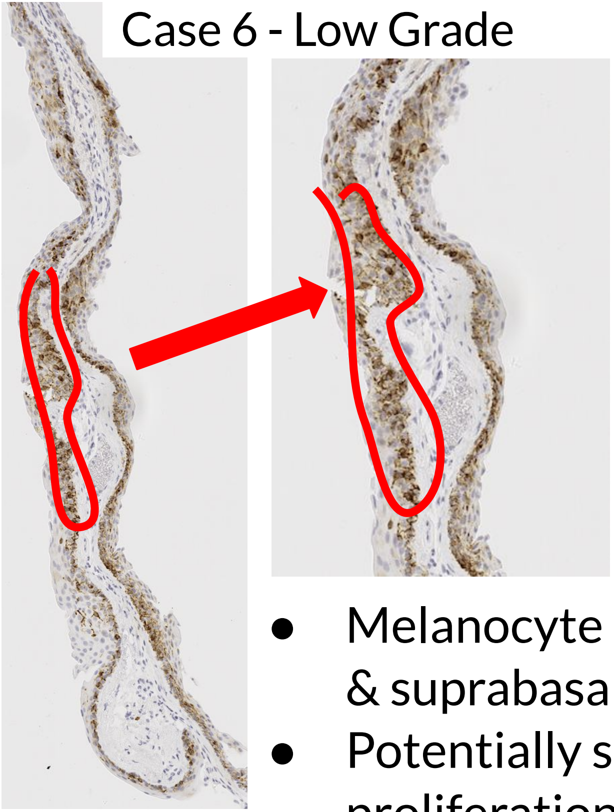




- Few brown spots (marking melanocyte cell bodies/cytoplasm) and diffuse brown patches indicate that the melanosis seen in H&E is not indicative of atypical melanocytic proliferation but instead benign pigment deposit
- <u>*Implementing a system of cross-referencing across the matched stains per case would make case 4 easier to classify by dispelling some of the concerns raised within the H&E stain analysis*</u>

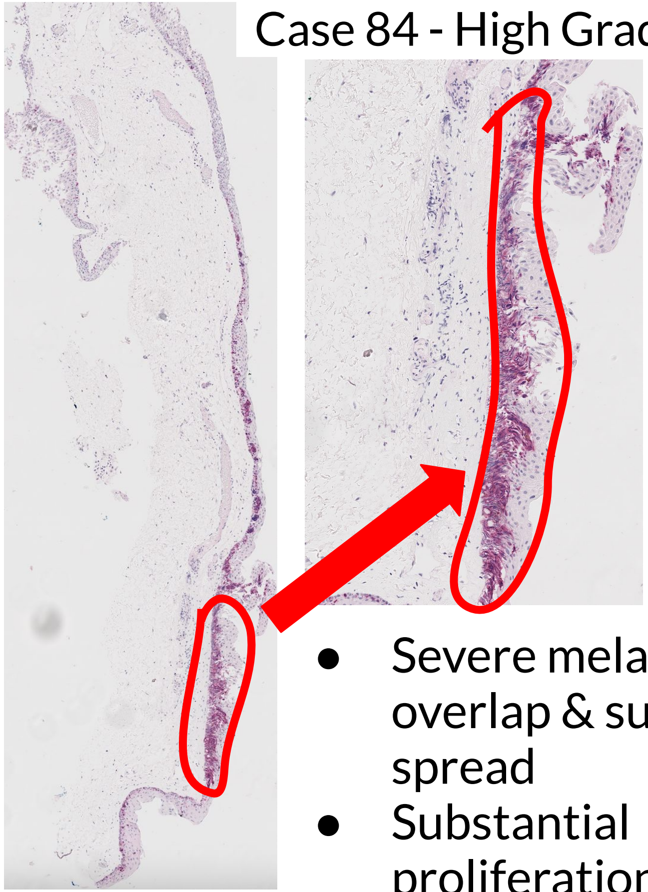
Example Low & High Grade *Melan-A* slices

Case 6 - Low Grade



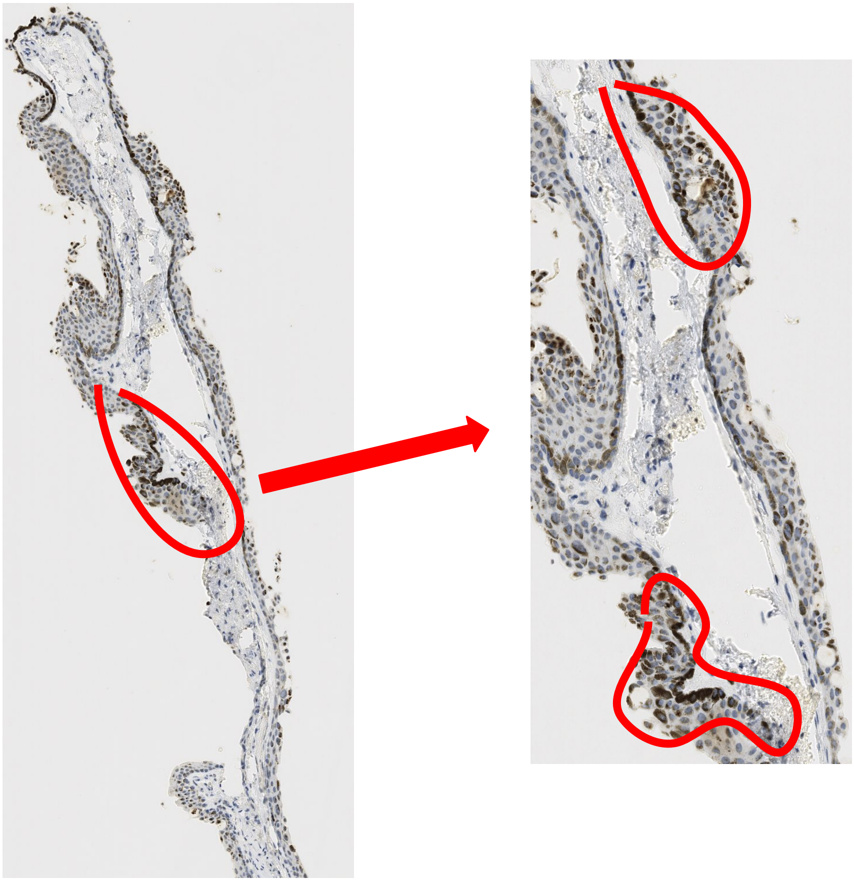
- Melanocyte overlap & suprabasal spread
- Potentially some proliferation

Case 84 - High Grade



- Severe melanocyte overlap & suprabasal spread
- Substantial proliferation

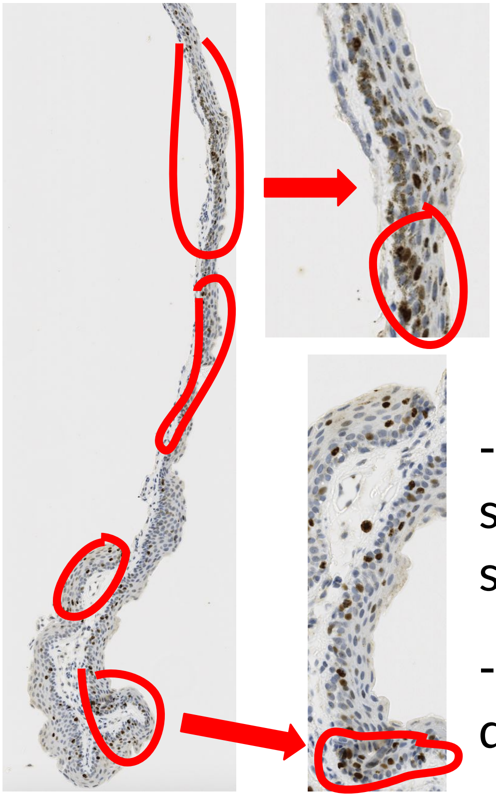
Case 4 *SOX10*



- Suprabasal spread is evident and some melanocytic clustering is present, albeit mostly at the basal layer which can be benign
- Cells within suprabasal spread are organized, small, round
    - Few have hyperchromatic nuclei
- Overall, this case is tough to classify, as indicated by the H&E stained slice

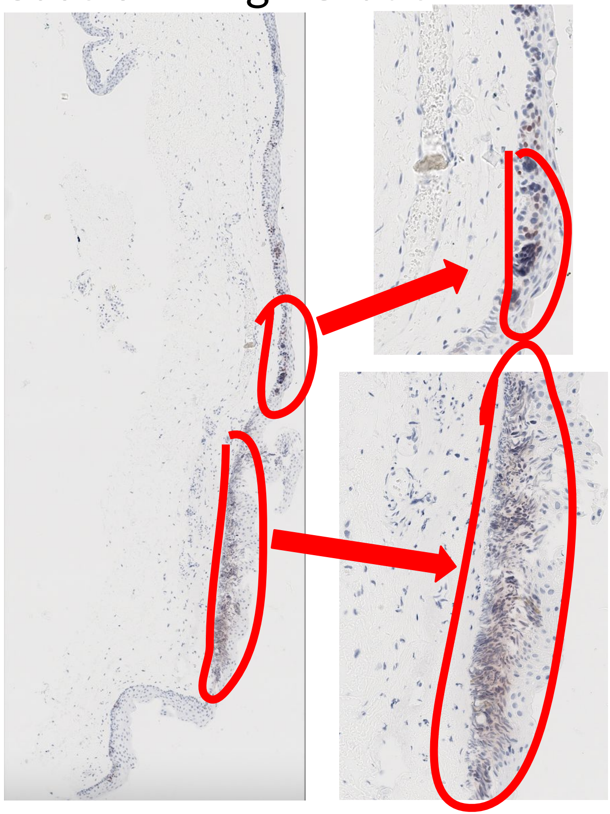
Example Low & High Grade *SOX10* slices

Case 6 - Low Grade



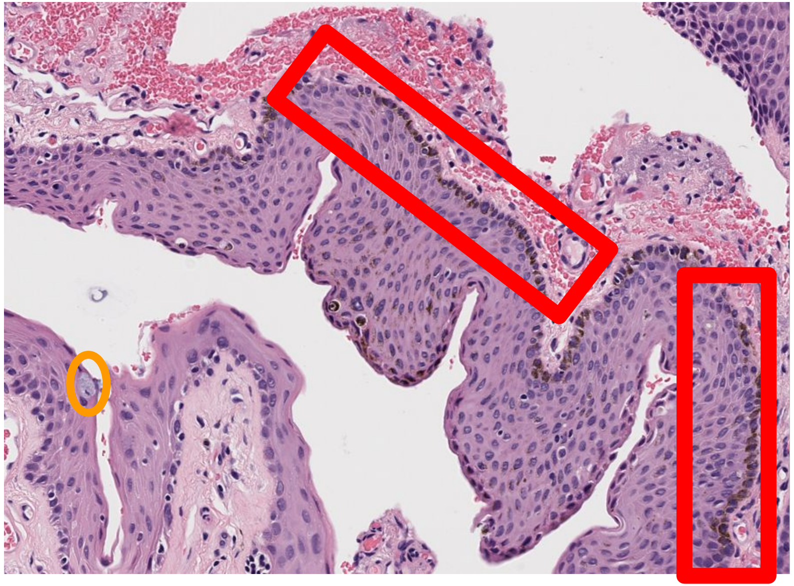
- Melanocytic suprabasal spread
- Disorganized distribution

Case 84 - High Grade

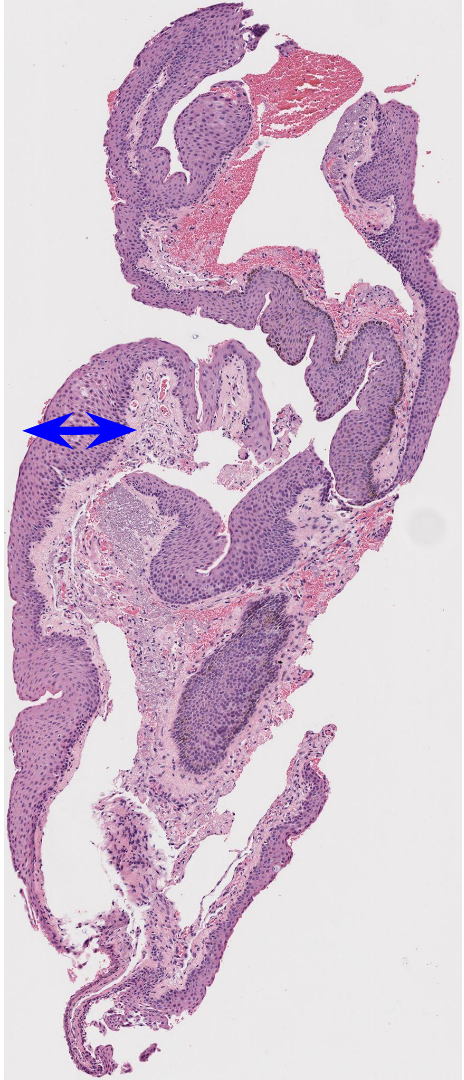


- Severe melanocytic suprabasal spread
- Melanocytic overlapping, proliferation

Case 21 - *benign*



- Few goblet cells - not hyperplasia
- Cysts - negligible presence
- Melanosis - slight brown pigment at basal layer of some part of the slice, not throughout

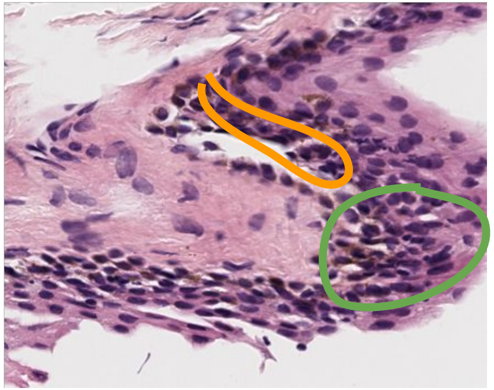


Medium to classify

- Maturation - benign; cells are small, organized, round
- Negligible signs of atypia - very few hyperchromatic nuclei

Case 22 - *benign*

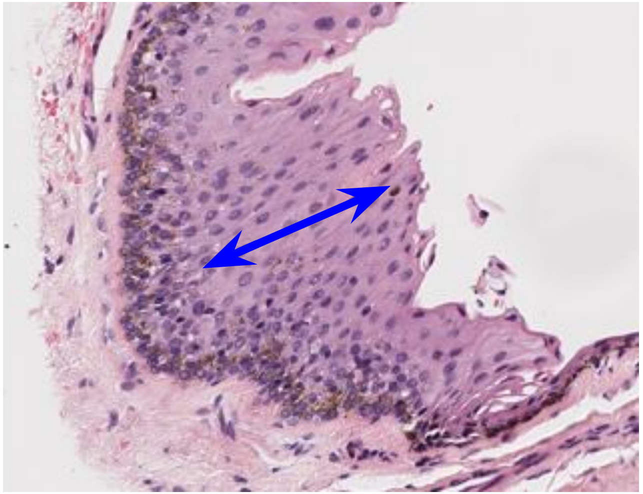
Hard to classify



← One area with lots of clustering

- Cysts present, benign indicator

- Few hyperchromatic nuclei along the basal layer



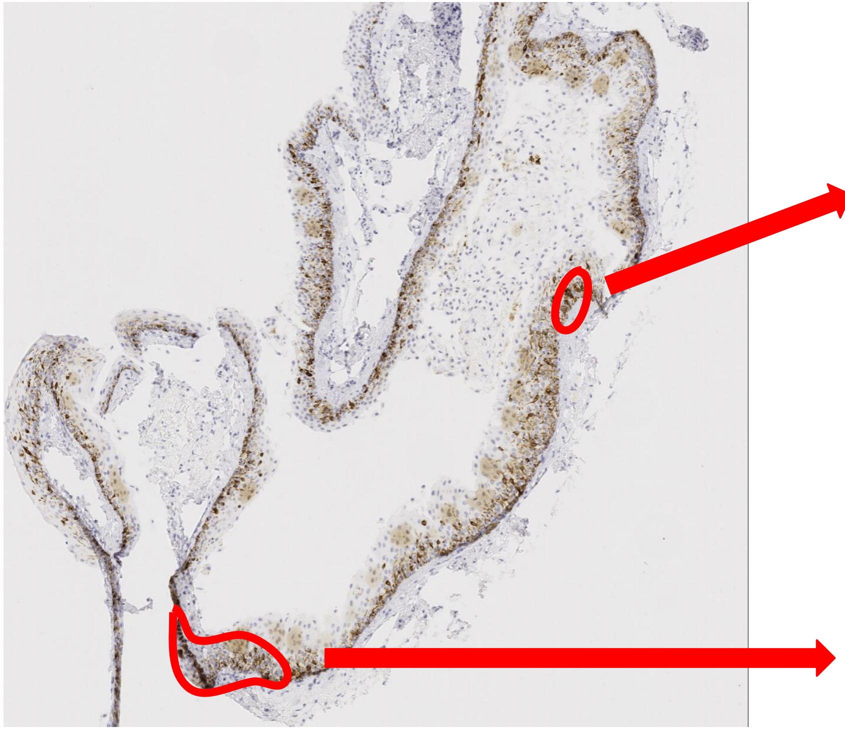
← Maturation mostly benign except for in the zoomed image above; cells are organized, small, round

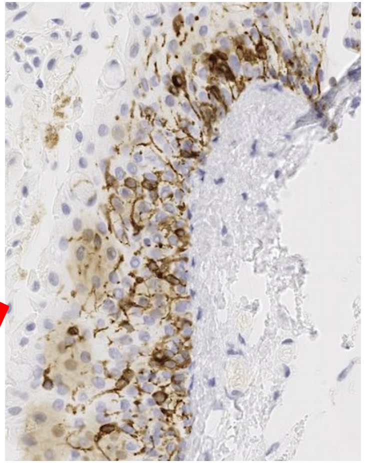
- Minimal goblet cell presence

- Negligible atypia outside of the zoomed image of clustering

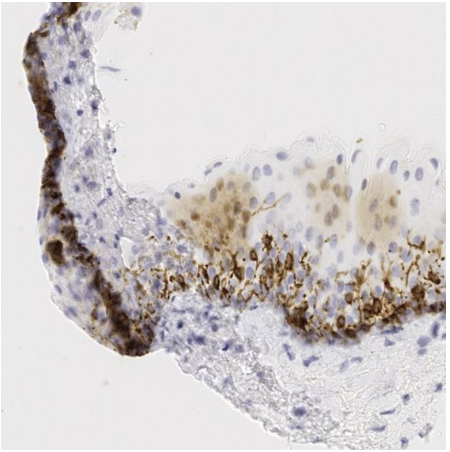
- Some light melanosis along basal layer, and a tiny bit of suprabasal spread

Case 22 *Melan-A*





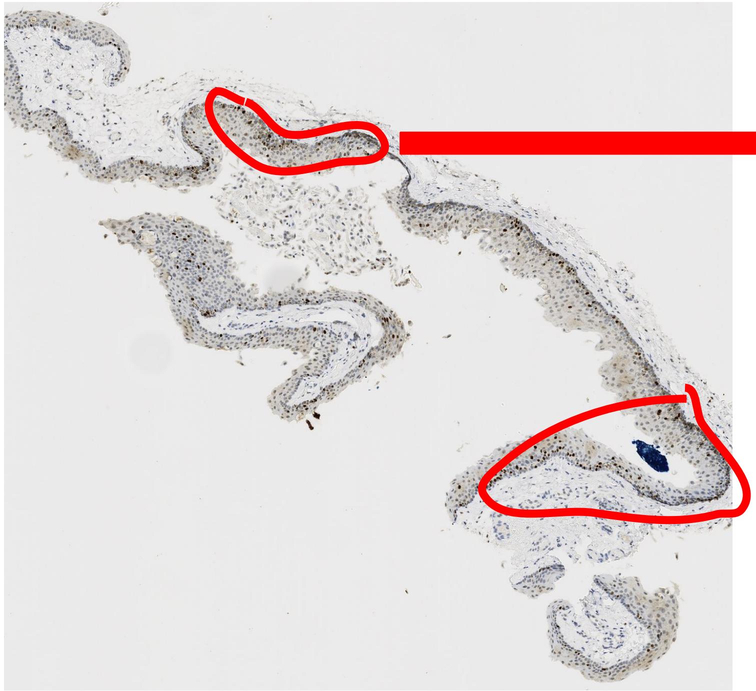
- Slight suprabasal spread of melanocytes
- No major overlapping or crowding

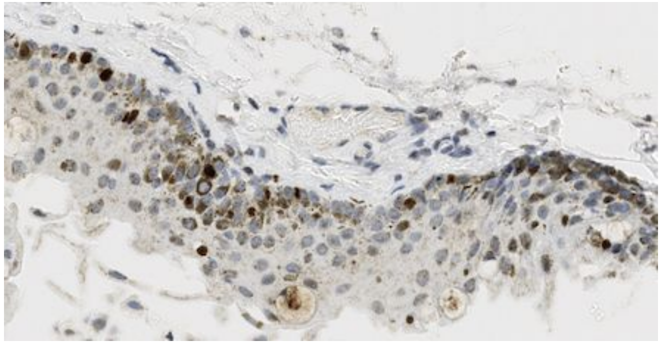


- Slight suprabasal spread
- Some melanocytic clustering

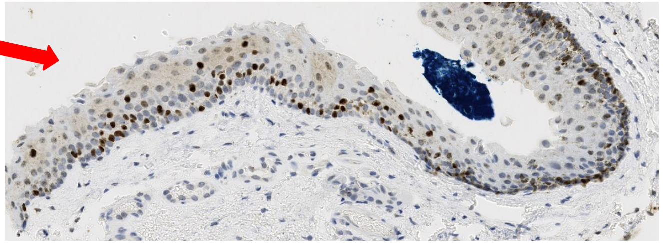
Diffuse staining spread throughout epithelium, but not marking melanocytes specifically – pigment contrast could be mistakenly flagged

Case 22 *SOX10*





- Slight suprabasal spread of melanocytes
- No major overlapping or crowding

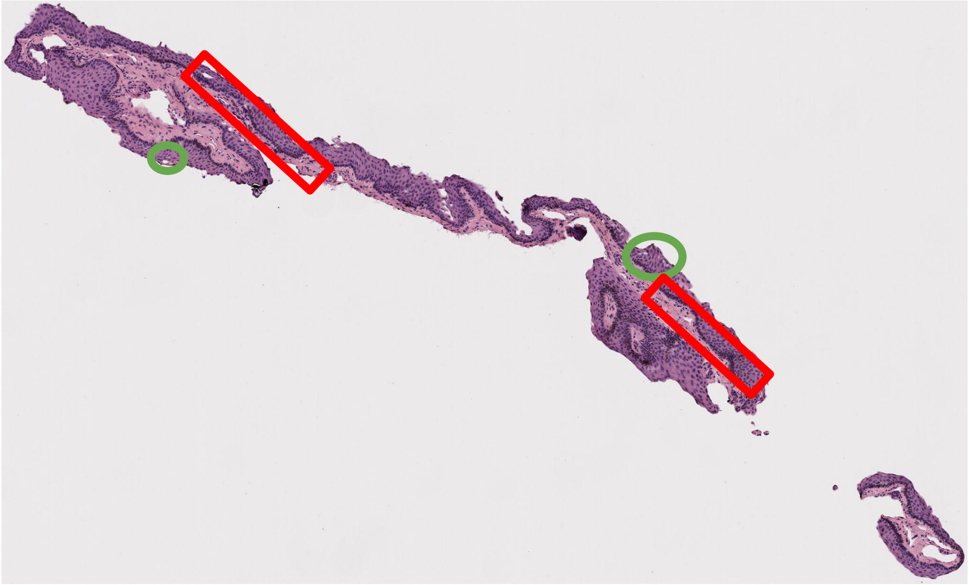


- Slight suprabasal spread
- Disorganized distribution of melanocytes

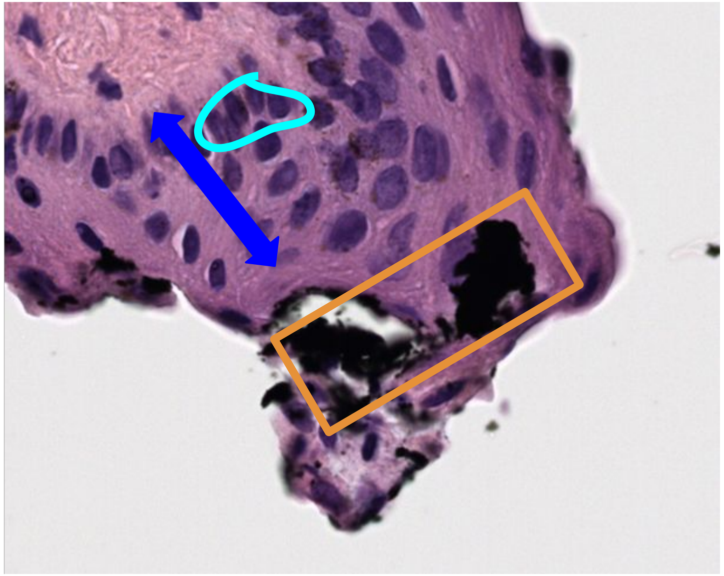
As in the Melan-A stained slice, there is some diffuse stain here throughout the epithelium, not specific to melanocytic nuclei

Case 23 - *benign*

Medium to classify



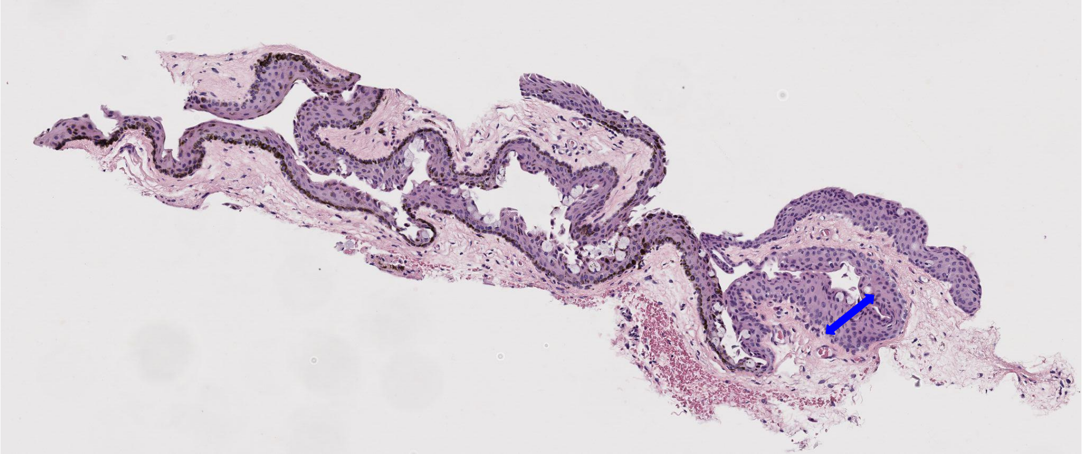
- Basal layer melanosis
- Cysts - benign indicator

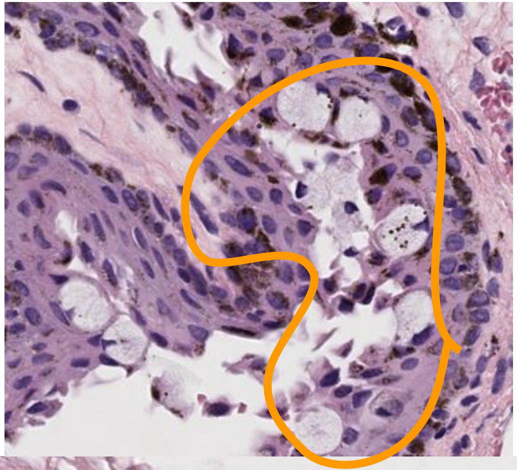


- Benign maturation (cells are organized, small, round)
- Potential imaging artifact
- Slight atypia (some hyperchromatic nuclei along basal layer, not throughout intraepithelial space)

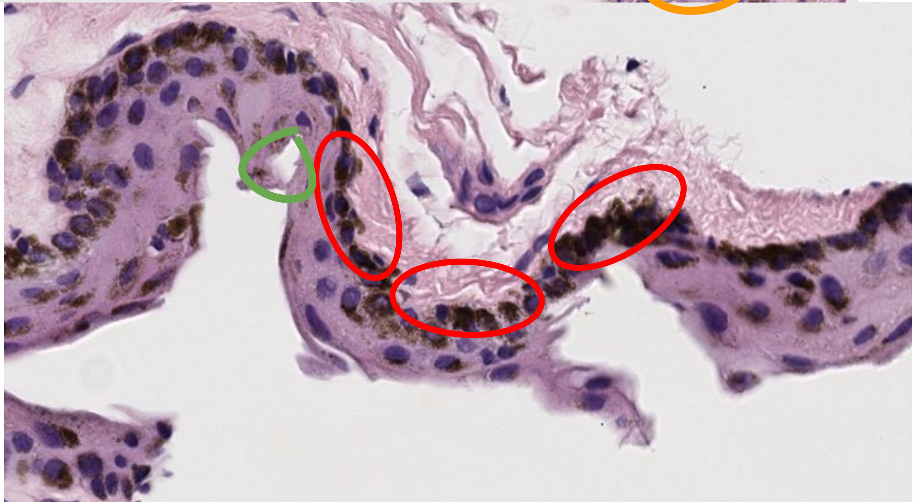
Case 24 - *benign*

Hard to classify

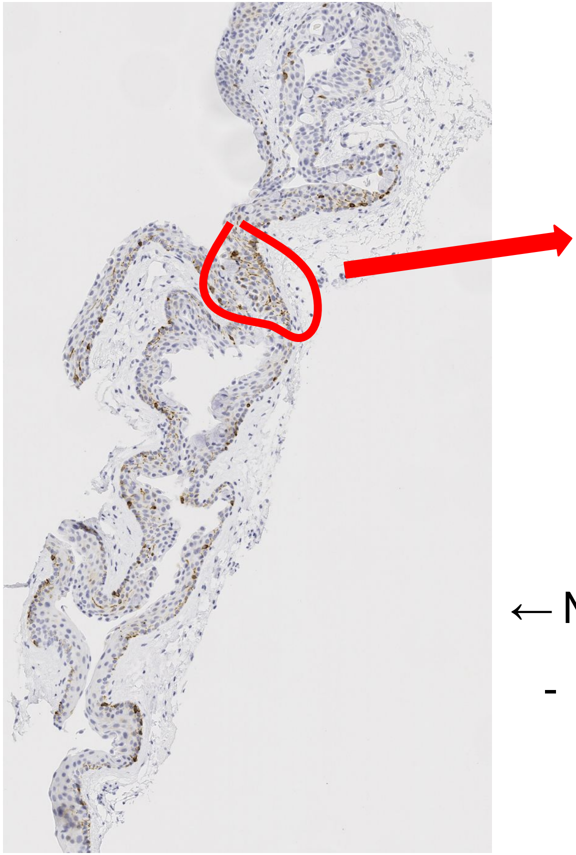


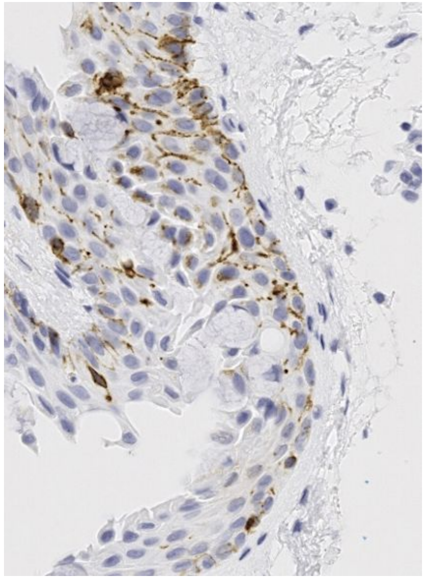


- Benign maturation (cells are organized, small, round)
- Goblet cell hyperplasia - benign
- No atypia (only some cells appear elongated)
- Pigmentation/melanosis throughout basal layer that also extends upward →
- Few cysts



Case 24 *Melan-A*



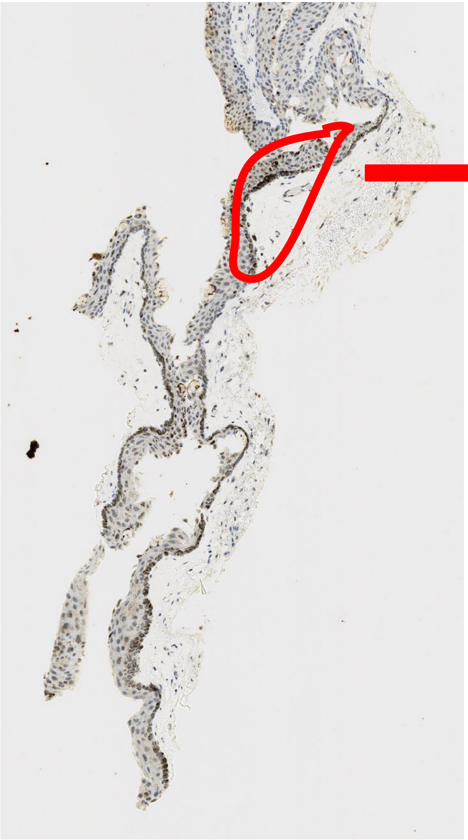


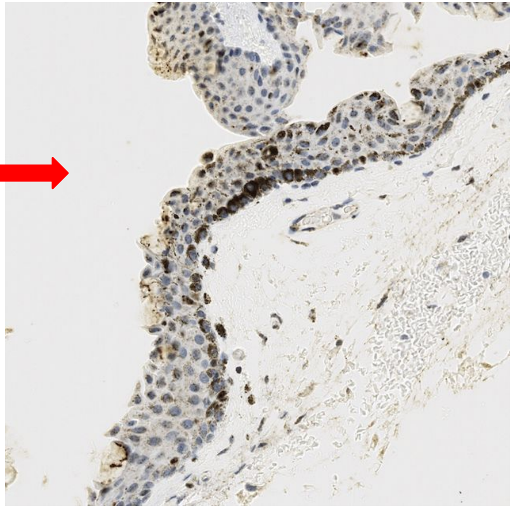
Zoom-in reinforces benign patterns present in this slice, clarifying and reducing some of the concerns posed by the H&E stained slice

← Minimal diffuse splotches of stain to impact model classification

- Melanocytes are primarily at the basal layer, and there is negligible overlap, crowding, or suprabasal spread
  - Melanosis seen in H&E is likely benign melanin deposits rather than atypical melanocytic proliferation

Case 24 *SOX10*





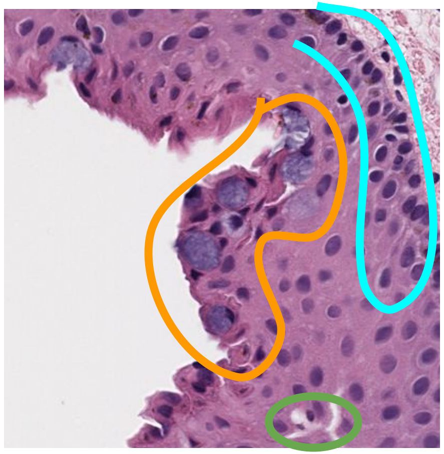
- Zoom-in shows the extent of the clustering and the suprabasal spread in the most severe part of the slice
    - Even here, spread is minimal and clustering does not turn into overlap

← Minimal diffuse splotches of stain to impact model classification

- Melanocytes are primarily at the basal layer, but there is some clustering in a few portions of the slice and some light suprabasal spread as well

Case 25 - *benign*

Hard to classify



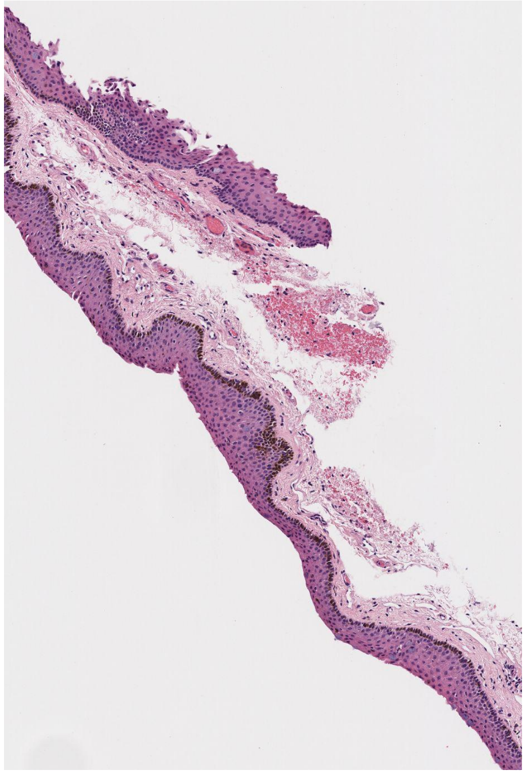
- Multiple goblet cell clusters present in this one region (very few other clusters in slice); not full hyperplasia

- Cysts present, benign indicator

- Thick basal & suprabasal layer of hyperchromatic nuclei

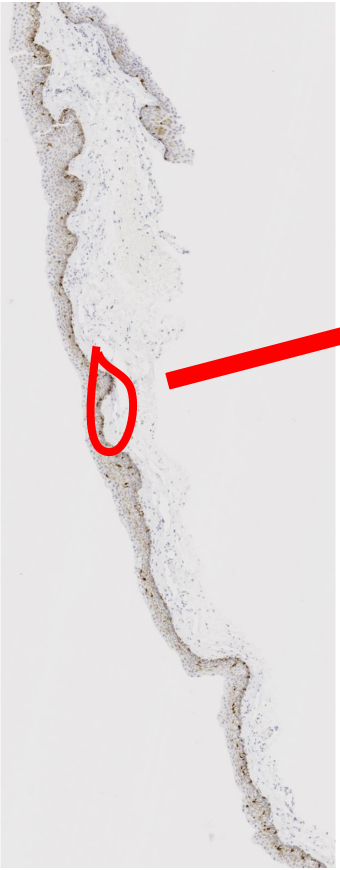
Melanosis at basal layer; more concentrated in some parts of the slice

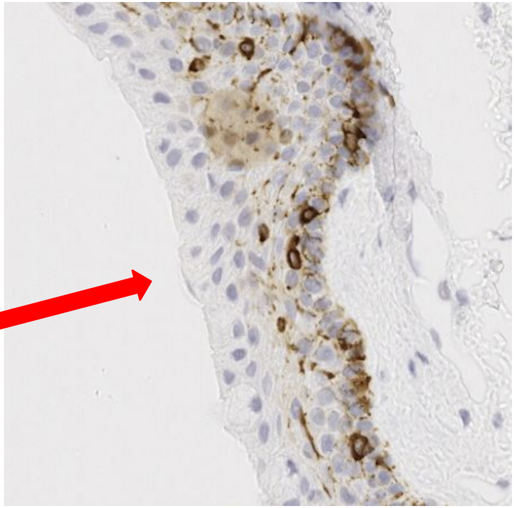
Maturation is benign; cells are small, round, organized



Full slice view reinforces the benign maturation alongside the presence of melanosis and hyperchromatic nuclei (darkness along basal layer)

Case 25 *Melan-A*



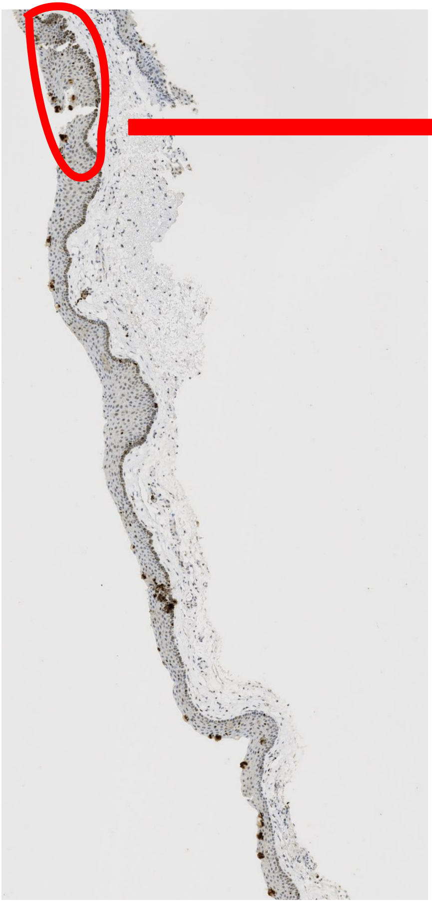


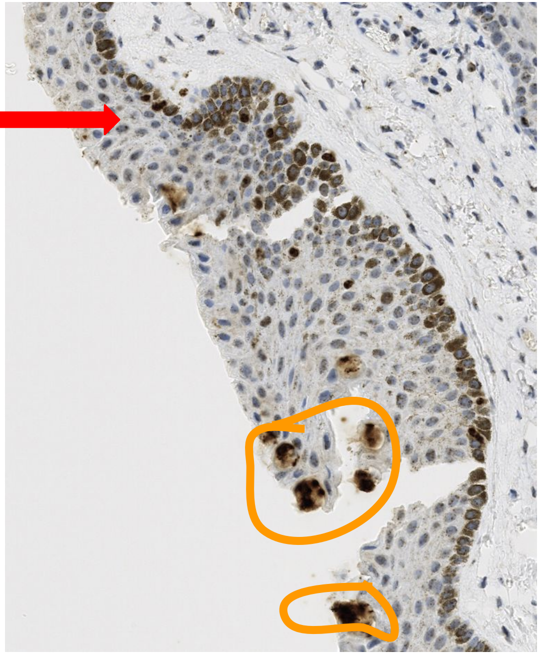
- Zoom-in reinforces the benign patterns present in this slice, clarifying and reducing some of the concerns posed by the H&E stained slice
- However, the slice also shows slight suprabasal spread, which could misconstrue classification

← Some diffuse splotches of stain, may impact model classification

- Melanocytes are primarily at the basal layer, without clustering, in a few portions of the slice
- Some light suprabasal spread is present

Case 25 *SOX10*





- Zoom-in shows the extent of the clustering and the suprabasal spread in one of the few severe parts of the slice
  - Clustering does not turn into overlap, but there are some big clusters near the top edge of the epithelium
    - It is unclear whether these are artifacts of staining or atypical melanocytes

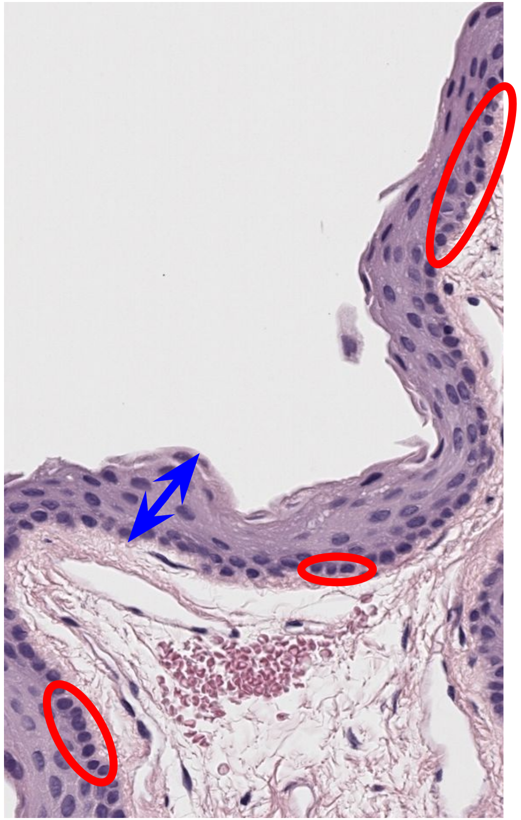
← Some, but fewer, diffuse splotches of stain to impact model classification

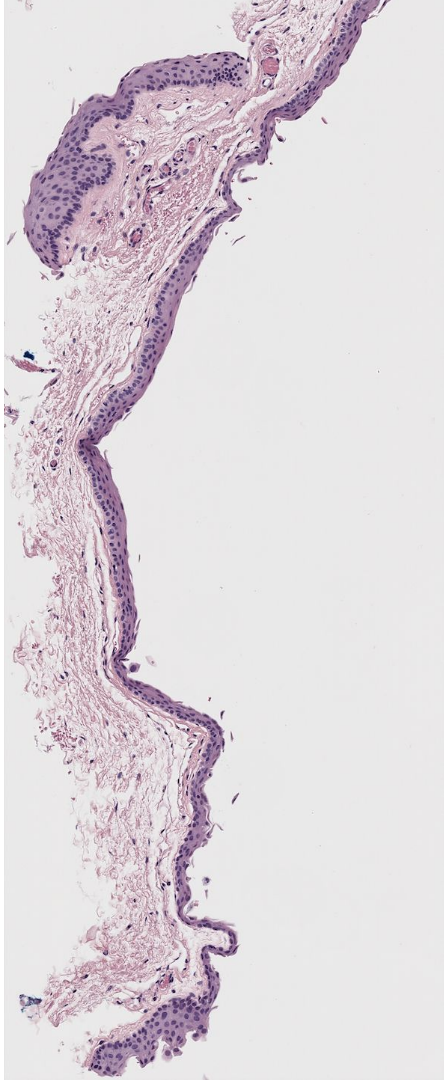
- Melanocytes are clustered at the basal layer, in many portions of the slice, and there is some suprabasal spread as well

Case 26 - *benign*

Easy to classify

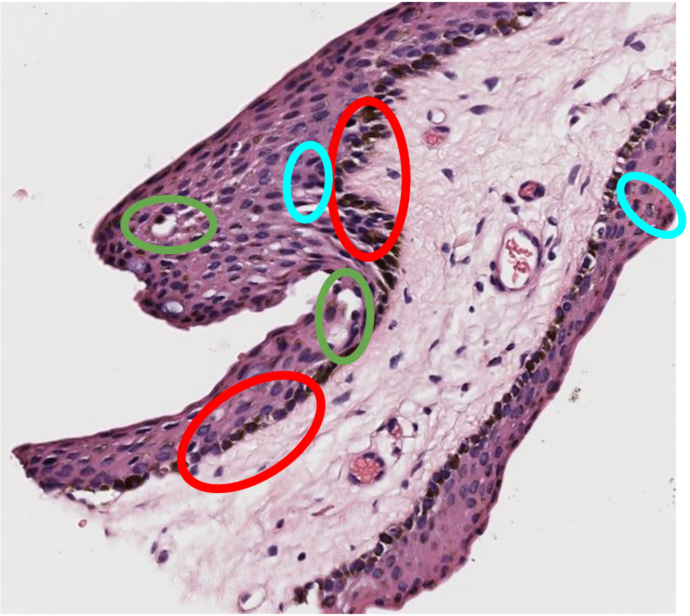
- Few hyperchromatic nuclei at basal layer; negligible atypia
- Benign maturation - cells are small, organized, round, and not overly clustered
- Minimal goblet cell presence - no hyperplasia
- No cysts





No melanosis present in any region of full slice

Case 27 - *benign*



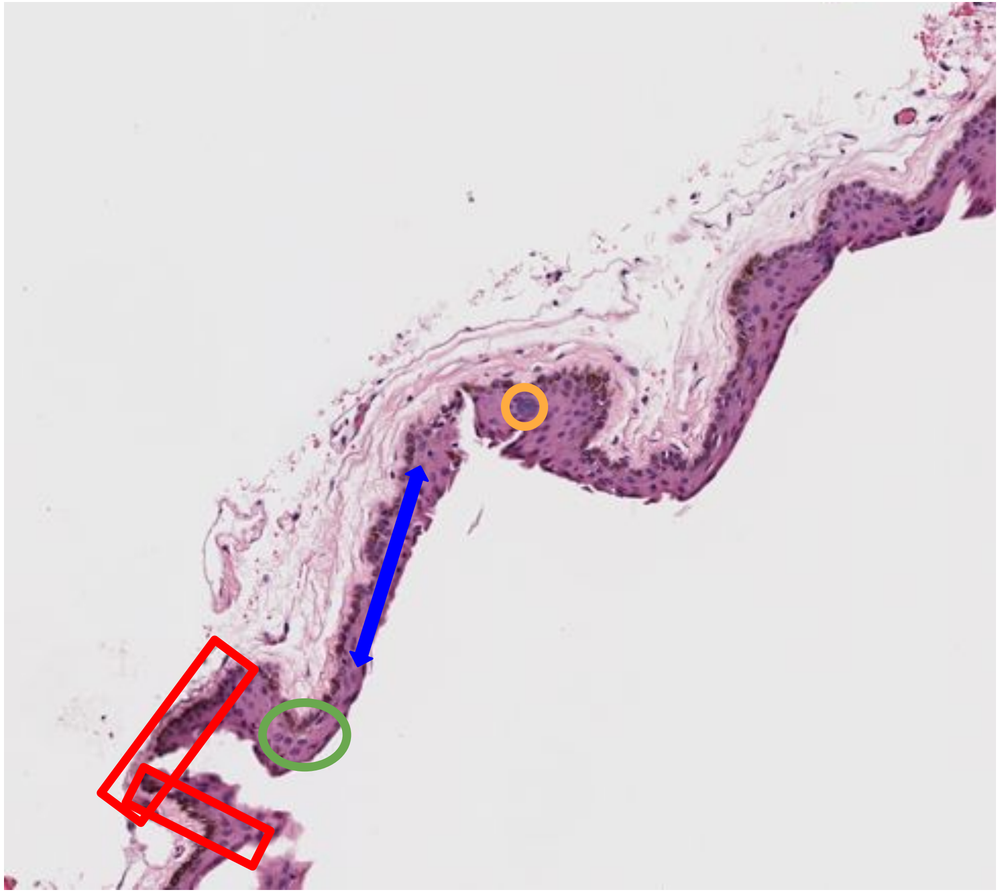
- No goblet cell hyperplasia
- Cysts present, benign indicator
- Mild melanosis and atypia/nuclear hyperchromasia concentrated near basement membrane on bottom half
- Maturation is largely benign; cells are small, round, organized

Medium to classify

Case 30 - *benign*

Medium to classify

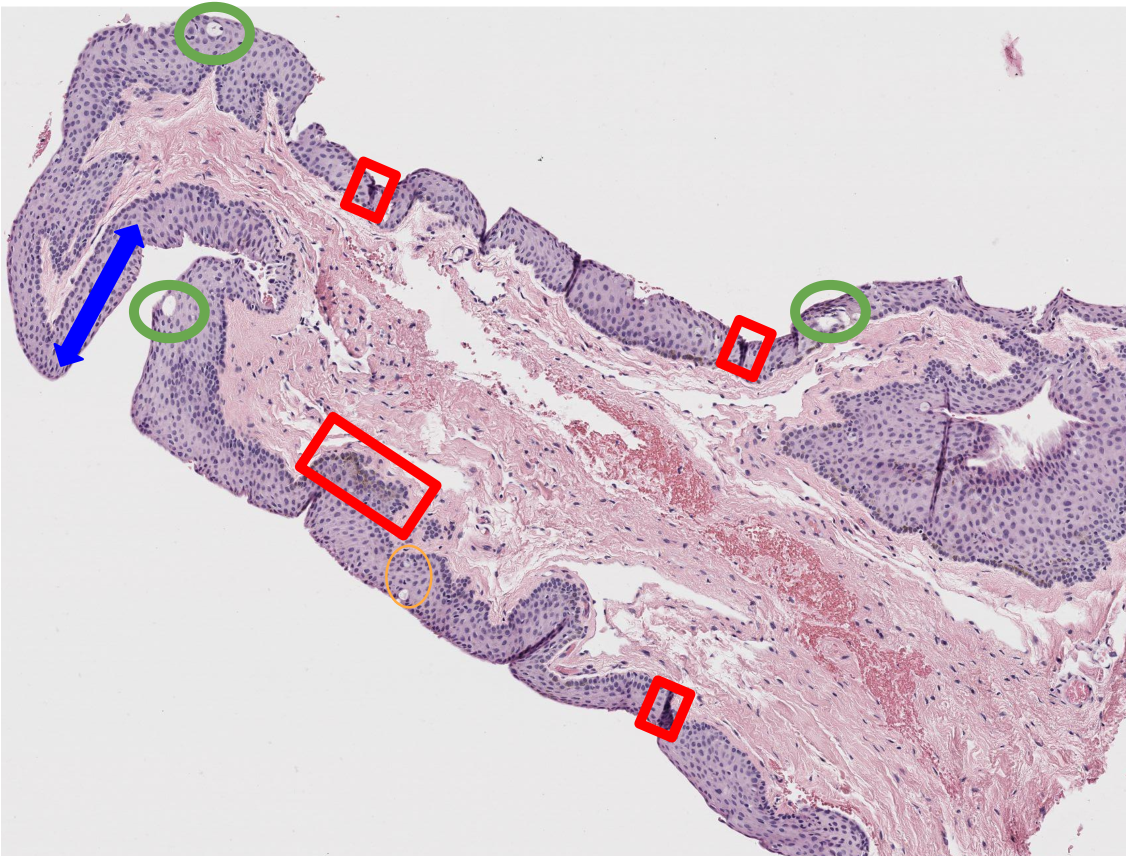
- A few goblet cells present
  - Might appear in some patches used, but not full hyperplasia
- A few cysts present
- Maturation resembles benign nevi
- Mild melanosis/hyperchromasia near basal layer on some parts of slice



Case 34 - *benign*

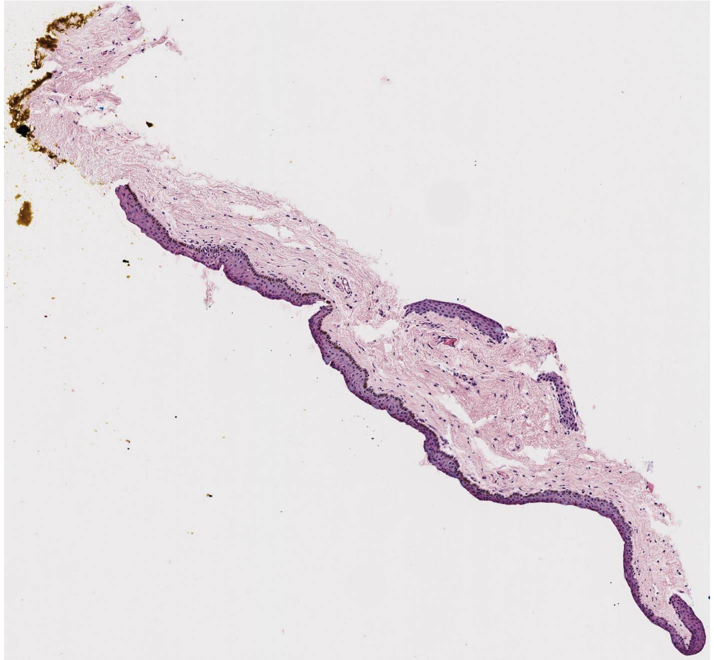
Easy to classify

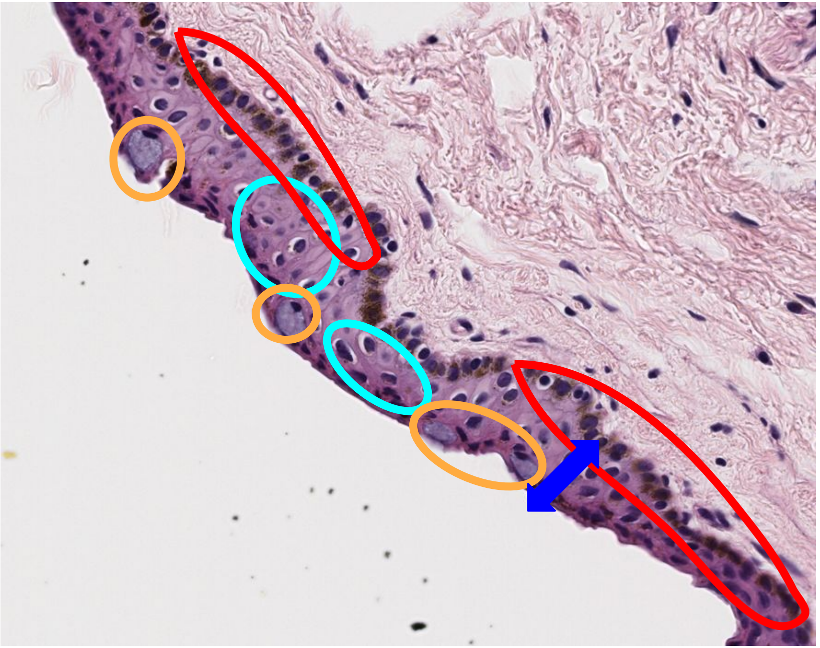
- A few goblet cells present
- A few cysts present
- Maturation resembles benign nevi
- Very slightly darker pigmentation/melanosis seen around a few small locations of basement membrane



Case 36 - *benign*

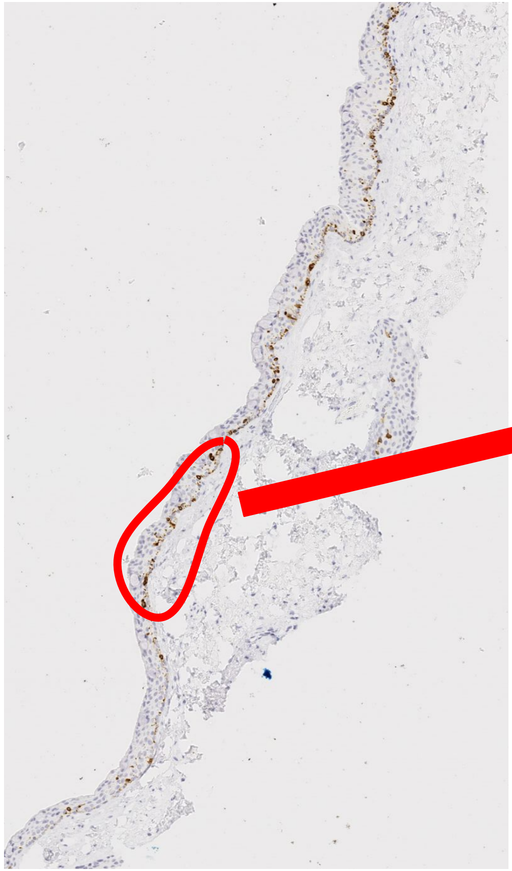
Hard to classify

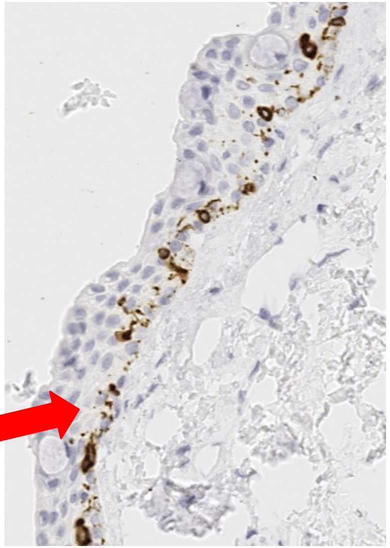




- Increased nuclear-cytoplasmic ratio, atypia (some nuclear enlargement & hyperchromatic cells throughout intraepithelial space)
- Goblet cells present in the pictured zoom, but no full hyperplasia in whole slice
- Maturation resembles benign nevi
- Melanosis present at basal layer

Case 36 *Melan-A*



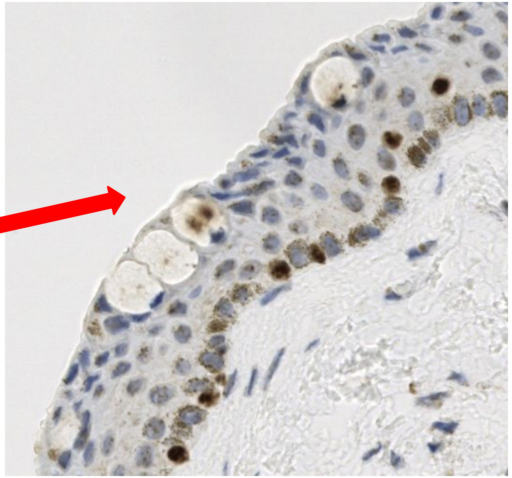


- Atypical cytoplasmic ratio and nuclear hyperchromasia is clarified by this zoom-in
    - Other cells (e.g. keratinocytes and other epithelial cells) are easy to confuse with melanocytes

← Negligible diffuse splotches of stain to impact model classification

- Melanocytes are almost entirely at the basal layer, and there is negligible overlap, crowding, or suprabasal spread
    - Slight melanosis seen in H&E is likely benign melanin deposits rather than atypical melanocytic proliferation

Case 36 *SOX10*



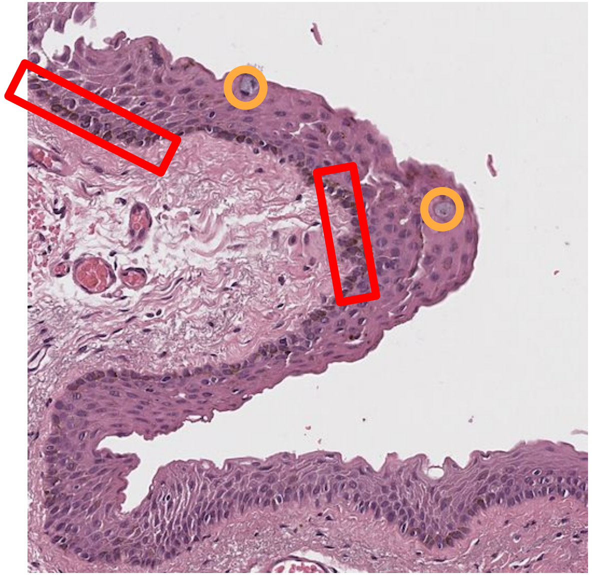
- Atypical cytoplasmic ratio and nuclear hyperchromasia is clarified by this zoom-in
    - Other cells (e.g. keratinocytes and other epithelial cells) are easy to confuse with melanocytes

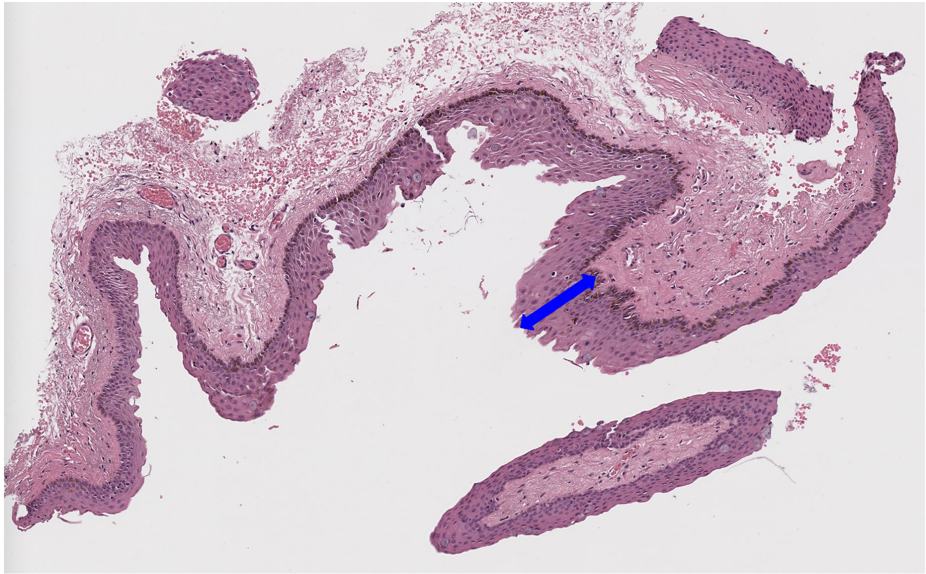
← Negligible diffuse splotches of stain to impact model classification

- Melanocytes are almost entirely at the basal layer, and there is negligible overlap, crowding, or suprabasal spread
    - Slight melanosis seen in H&E is likely benign melanin deposits rather than atypical melanocytic proliferation

Case 43 - *benign*

Easy to classify



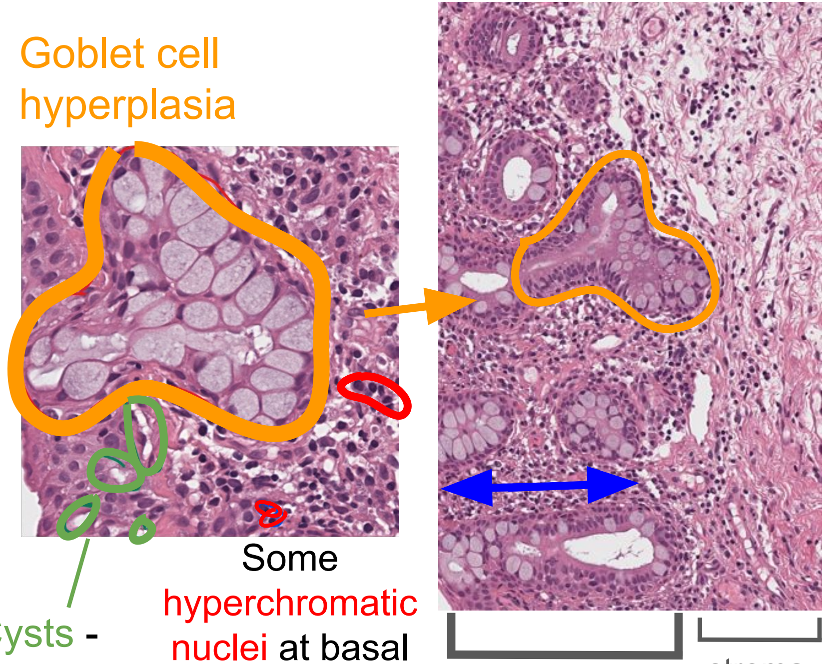


Benign
maturation

No cysts

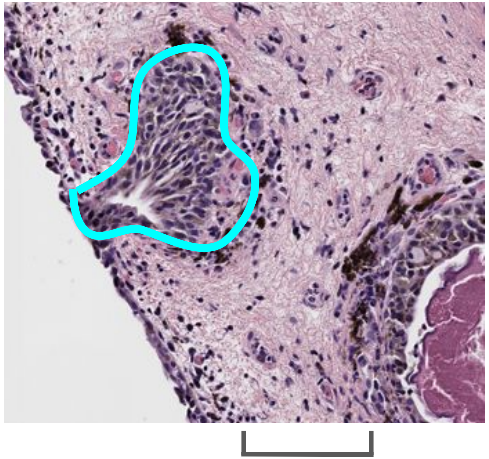
- A few goblet cells present but not hyperplasia
- No atypia
- Very slight melanosis at very few locations of basal layer/basement membrane, some slight inflammation

Case 46 - *benign*



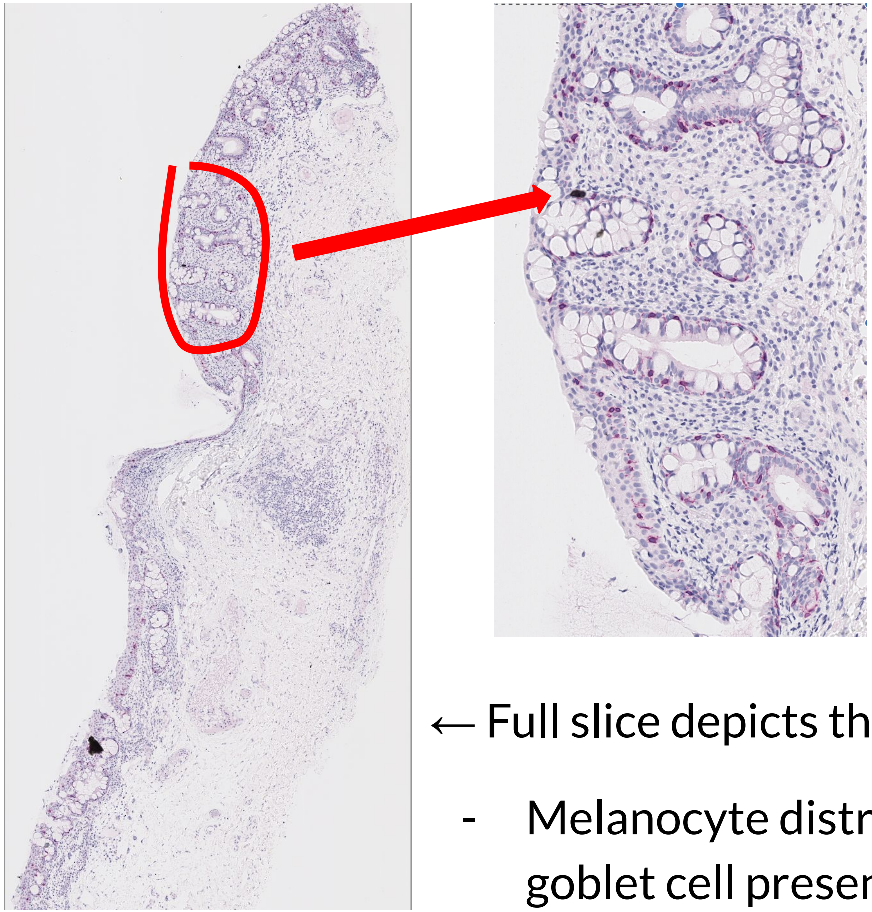
- Maturation appears <u>benign</u> - small, round, organized cell structure beyond goblet cell presence
- <u>No melanosis</u>

<u>Hard</u> to classify



- Case 3 (high grade) for comparison
  - Goblet cell clusters in oblong shapes (e.g. case 46) might be confused for melanocytes invading the stroma

Case 46 *Melan-A*

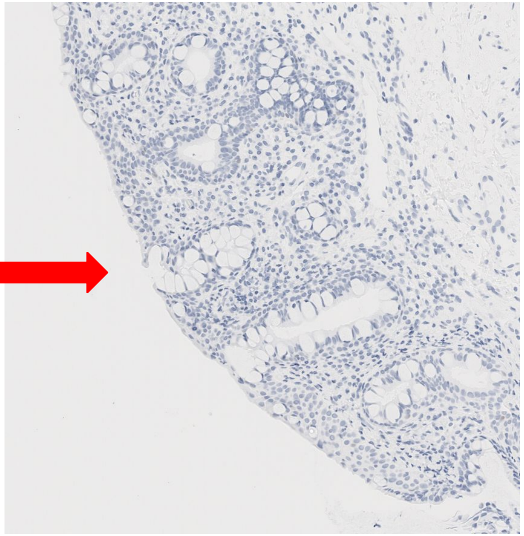


- Zoom-in reinforces how goblet cell hyperplasia is the main confounding factor in this case
    - Melanocyte distribution (in dark pink) appears disorganized and scattered because of goblet cell invagination

← Full slice depicts the large extent of goblet cell hyperplasia

- Melanocyte distribution appears disorganized because of goblet cell presence, potentially skewing model classification

Case 46 *SOX10*



- Zoom-in reinforces how goblet cell hyperplasia is the main confounding factor in this case
  - Melanocyte distribution (in dark pink) appears disorganized and scattered because of goblet cell invagination

← Full slice depicts the large extent of goblet cell hyperplasia

- Melanocyte distribution appears disorganized because of goblet cell presence, potentially skewing model classification

2.5 Takeaways

1. Most slices with comparatively more melanosis at the basal layer were initially misclassified by the Fall 2025 group
    a. We need to look at the current high-attention patches to see if this correlation is truly causing lower accuracy and determine which of the other “concerning” features discussed are biasing the model toward incorrect classifications
2. Slices with goblet cell invagination were often misclassified
    a. We think it might be getting confused for melanocytes invading the stroma, a classic finding for melanoma/high grade C-MIL
3. It is vital to combine information across different patches (or broadly across the different match slices), before classification or assigning any attention weights to make a holistic diagnosis; it is hard to make judgement using information from each technique separately (as the model currently does)

2.6 Recommendations

1. Region based attention weights
   a. Melanosis confined mostly to the basal layer is typically benign melanosis
      i. Sporadic, disorganized brown pigment in H&E stains that spreads suprabasally and more throughout the epithelium because of atypical cell maturation is a better indicator of the melanosis being atypical
   b. Same principle as above for nuclear hyperchromasia
   c. Look at cell clustering by region (more at basal layer is fine), but benign cases distinctly have more spacious spread in upper parts of epithelium whereas higher-grade cases have more clustering and overlap there
2. Ignore goblet cells (same recommendation as last week)
3. Ignore diffuse splotches of stain (mostly in Melan-A and SOX10 slices) that are not specific to cell bodies or nuclei

References

GenAI tools

Fall 2025 Presentation 4

Winter 2026 Presentation 6

Butt, K.; Hussain, R.; Coupland, S.E.; Krishna, Y. Conjunctival Melanoma: A Clinical Review and Update. Cancers 2024, 16, 3121. https://doi.org/10.3390/cancers16183121

https://www.pathologyoutlines.com/topic/eyepamconj.html#:~:text=Primary%20acquired%20melanosis%20(PAM)%20is%20also%20known,is%20significant%20factor%20for%20recurrence%20and%20progression

http://aao.org/eyenet/article/conjunctival-pigmented-lesions-diagnosis-managemen#:~:text=Complexion-Associated%20Melanosis-,Presentation.,pigmentation%20appears%20flat%20and%20noncystic.

https://onlinelibrary.wiley.com/doi/10.1111/aos.13100

https://www.spandidos-publications.com/10.3892/br.2022.1507#

https://entokey.com/pathology-of-conjunctiva/#:~:text=Bulbar:%20The%20bulbar%20conjunctiva%20extends,3

https://www.sciencedirect.com/science/article/pii/S1361841521000785
https://medicine.nus.edu.sg/pathweb/normal-histology/eye-conjunctiva/

References Contd.

https://pmc.ncbi.nlm.nih.gov/articles/PMC7583058/#:~:text=Background,not%20been%20histopathologically%20confirmed%20before.

https://pmc.ncbi.nlm.nih.gov/articles/PMC11429624/#:~:text=The%202022%20WHO%20classification%20of,%2DMIL)%20%5B49%5D.&text=Conjunctival%20hypermelanosis%20(increased%20pigment%20in,inconspicuous%20cytoplasm)%20may%20be%20seen.&text=Predominantly%20basilar%20melanocytic%20proliferation%20with,and%20inconspicuous%20or%20scant%20cytoplasm).&text=More%20confluent%20basilar%20and%20significant,growth%20and%20epithelioid%20cell%20cytomorphology.&text=The%20term%20melanoma%20in%20situ,documented%20evidence%20of%20subepithelial%20invasion.

https://pmc.ncbi.nlm.nih.gov/articles/PMC11429624/

Winter 2025 Presentation 3

Winter 2025 Presentation 4

Winter 2025 Presentation 5